

Network Theory

Previous Year Questions

GATE INSTRUMENTATION ENGINEERING

(IN)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Network Theory (IN), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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- AIR 27 — GATE (ECE)
- AIR 45 — GATE (IN)
- 5+ years of teaching experience



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- AIR 19 — GATE (IN)
- AIR 66 — GATE (EE)
- 3+ years in teaching & the VLSI industry



> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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4	2nd Order Dynamics and Complex Impedance	18
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UNIT

I

Basic Concepts of Networks

Topics

1. Introduction to Electrical Quantities
2. Basic Laws and Classification of Network Elements
3. Resistors, Capacitors and Inductors
4. Energy Sources in Electrical Circuits
5. Series-Parallel Connections
6. Methods of Analyzing Electrical Circuits — Nodal and Mesh Analysis

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Unit Snapshot*

Stream: IN	**Questions:** 28	**MCQ:** 22	**MSQ:** 0	**NAT:** 6
1M: 14	**2M:** 14	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2023)–4M(2015)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 MCQ 2026 2M

In the circuit shown below, the value of the output voltage V_0 is _____ V.

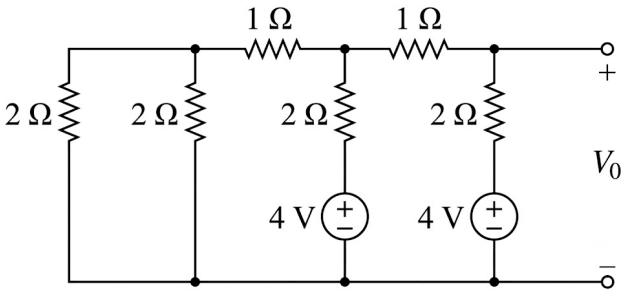


Fig. Q1.26.1

A. 2 **B.** 3 **C.** 4 **D.** 8

Q1.25.1 MCQ 2025 2M

The bridge circuit, shown in Figure (a), can be equivalently represented using the circuit shown in Figure (b). The values of R_1, R_2, and V_C in the equivalent circuit are

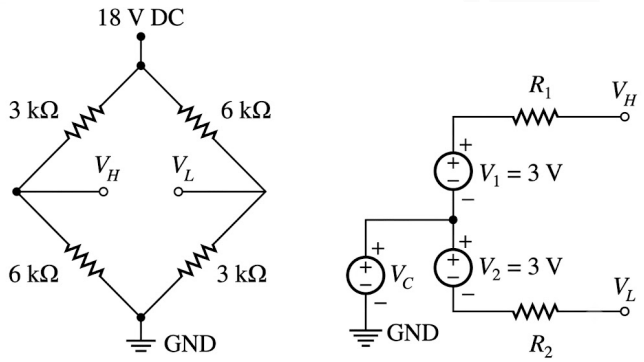


Fig. Q1.25.1

A. $R_1 = 6\,\text{k}\Omega$, $R_2 = 3\,\text{k}\Omega$ and $V_C = 9$ V

B. $R_1 = 3\,\text{k}\Omega$, $R_2 = 6\,\text{k}\Omega$ and $V_C = 4.5$ V

C. $R_1 = 2\,\text{k}\Omega$, $R_2 = 2\,\text{k}\Omega$ and $V_C = 9$ V

D. $R_1 = 2\,\text{k}\Omega$, $R_2 = 2\,\text{k}\Omega$ and $V_C = 4.5$ V

Q1.23.1 NAT 2023 1M

When the bridge given below is balanced, the current through the resistor R_a is _____ mA (rounded off to two decimal places).

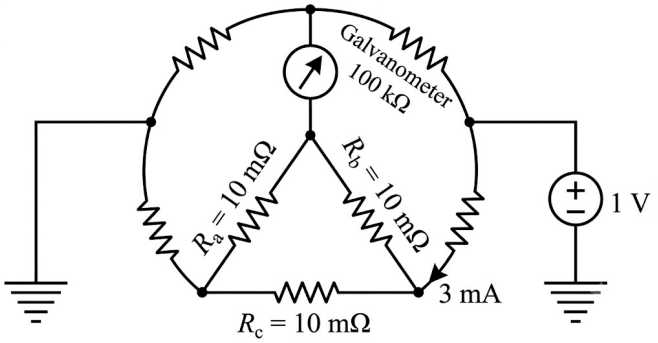


Fig. Q1.23.1

Q1.20.1 MCQ 2020 1M

In the circuit shown below, the safe maximum value for current I is

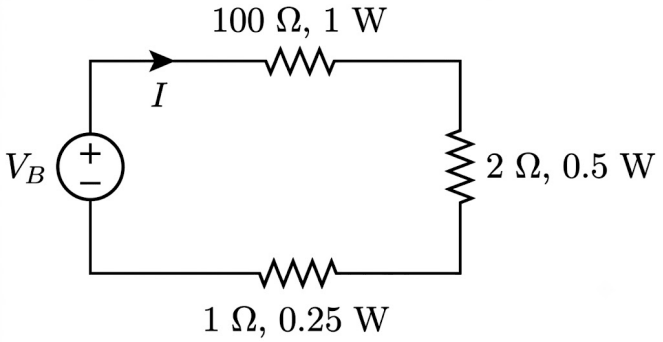


Fig. Q1.20.1

A. 0.1 A **B.** 1.0 A **C.** 0.5 A **D.** 0.05 A

Q1.20.2 MCQ 2020 2M

I_1, I_2 and I_3 in the figure below are mesh currents. The correct set of mesh equations for these currents, in matrix form, is

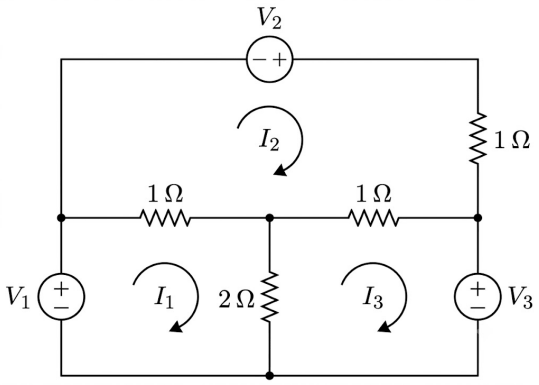


Fig. Q1.20.2

A. $\begin{bmatrix} -3 & -1 & -2 \\ -1 & 3 & -1 \\ -2 & -1 & 3 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ -V_3 \end{bmatrix}$

B. $\begin{bmatrix} 1 & -1 & -2 \\ -1 & 2 & -1 \\ -2 & -1 & 3 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}$

C. $\begin{bmatrix} 3 & -1 & -2 \\ -1 & 3 & -1 \\ -2 & -1 & 3 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ -V_3 \end{bmatrix}$

D. $\begin{bmatrix} 3 & -1 & -2 \\ -1 & 3 & -1 \\ -2 & -1 & -3 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}$

Q1.19.1 NAT 2019 1M

Consider a circuit comprising only resistors with constant resistance and ideal independent DC voltage sources. If all the resistances are scaled down by a factor 10, and all source voltages are scaled up by a factor 10, the power dissipated in the circuit scales up by a factor of _____.

Q1.18.1 MCQ 2018 2M

In the given circuit, the mesh currents I_1, I_2 and I_3 are

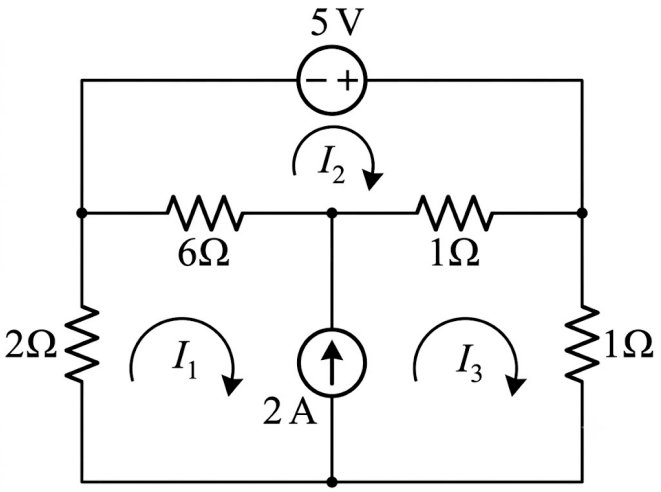


Fig. Q1.18.1

A. $I_1 = 1$ A, $I_2 = 2$ A and $I_3 = 3$ A

B. $I_1 = 2$ A, $I_2 = 3$ A and $I_3 = 4$ A

C. $I_1 = 3$ A, $I_2 = 4$ A and $I_3 = 5$ A

D. $I_1 = 4$ A, $I_2 = 5$ A and $I_3 = 6$ A

Q1.17.1 NAT 2017 1M

A circuit consisting of dependent and independent source is shown in the figure. If the voltage at Node-1 is −1 V, then the voltage at Node-2 is _____ V.

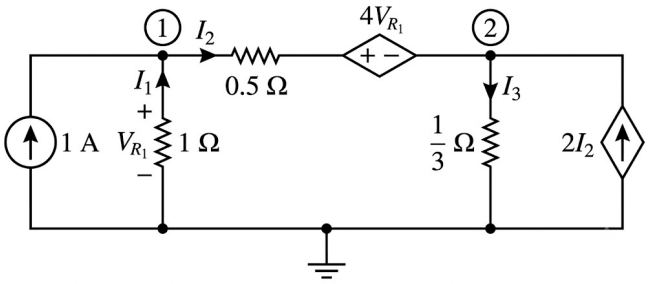


Fig. Q1.17.1

Q1.16.1 NAT 2016 2M

The current I_x in the circuit given below in milli-ampere is _____.

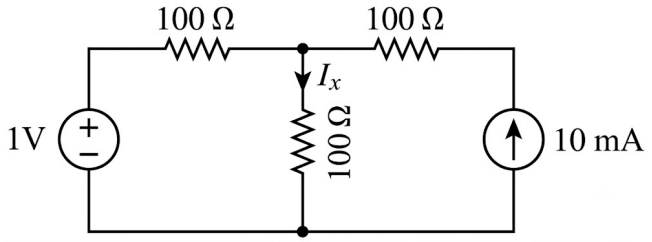


Fig. Q1.16.1

Q1.15.1 NAT 2015 2M

The current in amperes through the resistor R in the circuit shown in the figure is _____ A.

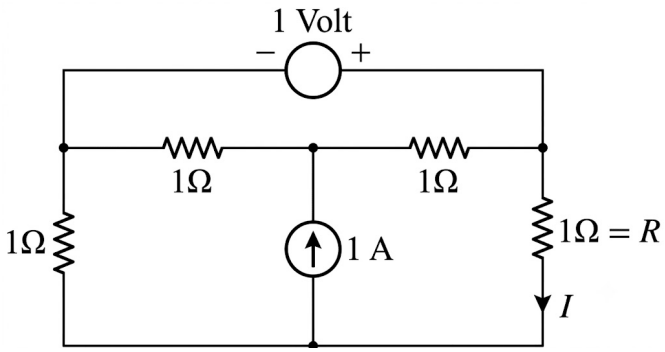
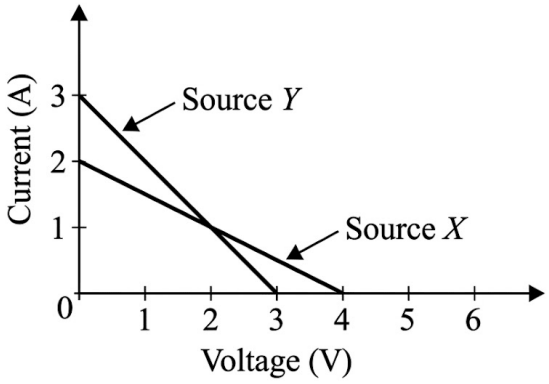


Fig. Q1.15.1

Q1.15.2 NAT 2015 2M

The linear I-V characteristic of 2-terminal non-ideal dc sources X and Y are shown in the figure. If the sources are connected to a 1 Ω resistor as shown, the current through the resistor in amperes is _____ A.



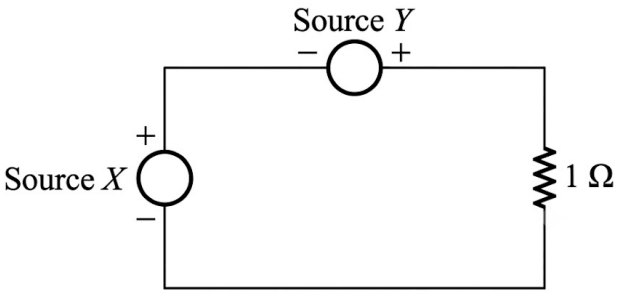


Fig. Q1.15.2

Q1.13.1 MCQ 2013 1M

Consider a delta connection of resistors and its equivalent star connection is shown below. If all elements of the delta connection are scaled by a factor k, $k > 0$, the elements of the corresponding star equivalent will be scaled by a factor of

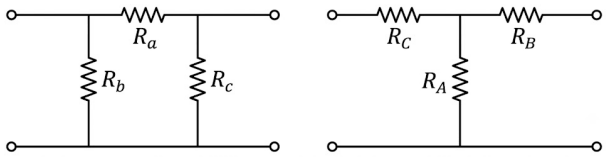


Fig. Q1.13.1

A. k^2 **B.** k **C.** $1/k$ **D.** $\sqrt{k}$

Q1.13.2 MCQ 2013 2M

Three capacitors C_1, C_2 and C_3 whose values are 10 μF, 5 μF and 2 μF respectively, have breakdown voltages of 10 V, 5 V and 2 V respectively. For the interconnection shown below, the maximum safe voltage in volts that can be applied across the combination, and the corresponding total charge in μC stored in the effective capacitance across the terminals are respectively,

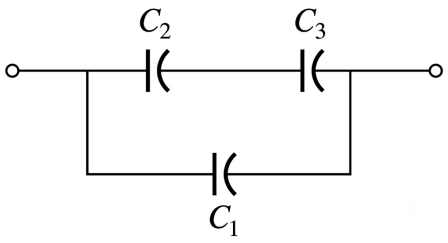


Fig. Q1.13.2

A. 2.8 and 36 **B.** 7 and 119
C. 2.8 and 32 **D.** 7 and 80

Q1.12.1 MCQ 2012 2M

If $V_A - V_B = 6$ V, then $V_C - V_D$ is

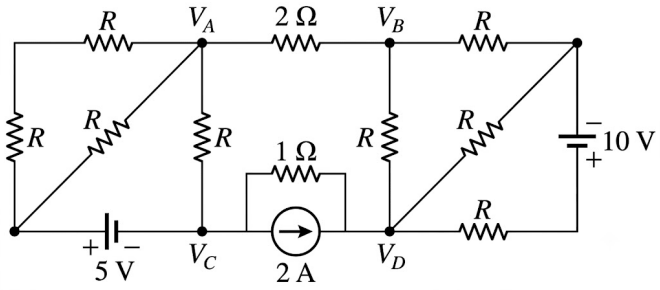


Fig. Q1.12.1

A. −5 V **B.** 2 V **C.** 3 V **D.** 6 V

Q1.11.1 MCQ 2011 1M

The current I shown in the circuit given below is equal to

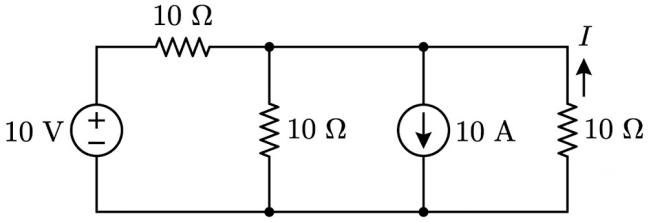


Fig. Q1.11.1

A. 3 A **B.** 3.67 A **C.** 6 A **D.** 9 A

Q1.10.1 MCQ 2010 1M

A 100 Ω, 1 W resistor and a 800 Ω, 2 W resistor are connected in series. The maximum dc voltage that can be applied continuously to the series circuit without exceeding the power limit of any of the resistors is

A. 90 V **B.** 50 V **C.** 45 V **D.** 40 V

Q1.09.1 MCQ 2009 2M

The root mean squared value of $x(t) = 3 + 2\sin(t)\cos(2t)$ is

A. $\sqrt{3}$ **B.** $\sqrt{8}$ **C.** $\sqrt{10}$ **D.** $\sqrt{11}$

Q1.08.1 MCQ 2008 2M

Which one of the following equation is valid for the circuit shown below?

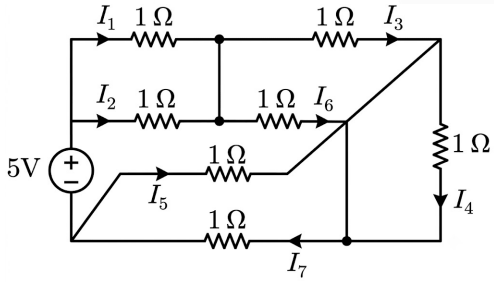


Fig. Q1.08.1

A. $I_2 + I_5 - I_6 + I_7 = 0$ **B.** $I_3 - I_5 + I_6 + I_7 = 0$
C. $I_3 + I_5 + I_6 + I_7 = 0$ **D.** $I_3 + I_5 + I_6 - I_7 = 0$

Q1.08.2 MCQ 2008 1M

The power supplied by the dc voltage source in the circuit shown below is

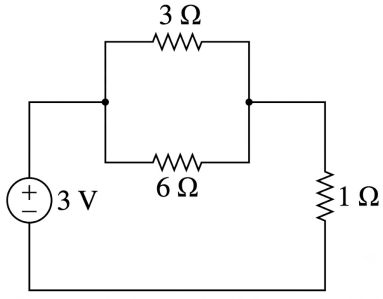


Fig. Q1.08.2

A. 0 W **B.** 1.0 W **C.** 2.5 W **D.** 3.0 W

Q1.08.3 MCQ 2008 1M

The current I supplied by the dc voltage source in the circuit shown below is

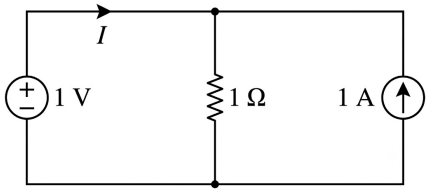


Fig. Q1.08.3

A. 0 A **B.** 0.5 A **C.** 1 A **D.** 2 A

Q1.08.4 MCQ 2008 2M

For the circuit shown below, the voltage across the capacitor is

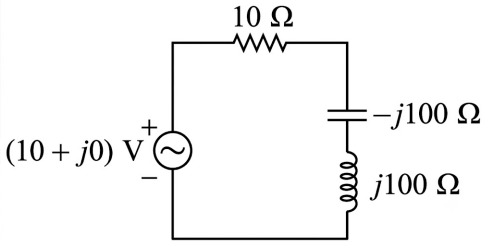


Fig. Q1.08.4

A. $(10 + j0)$ V **B.** $(100 + j0)$ V
C. $(0 + j100)$ V **D.** $(0 - j100)$ V

Q1.06.1 MCQ 2006 2M

A metal wire has a uniform cross-section A, length l, and a resistance R between its two end points. It is uniformly stretched so that its length becomes αl. The new resistance is

A. αR **B.** $\alpha^2 R$ **C.** $\sqrt{\alpha}\, R$ **D.** $e^{\alpha} R$

Q1.06.2 MCQ 2006 1M

The root-mean-square value of a voltage waveform consisting of a superposition of 2 V dc and a 4 V peak-to-peak square wave is

A. 2 V **B.** $\sqrt{6}$ V **C.** $\sqrt{8}$ V **D.** $\sqrt{12}$ V

Q1.04.1 MCQ 2004 2M

For the circuit shown in the figure, the diode D is ideal. The power dissipated by the 300 Ω resistor is

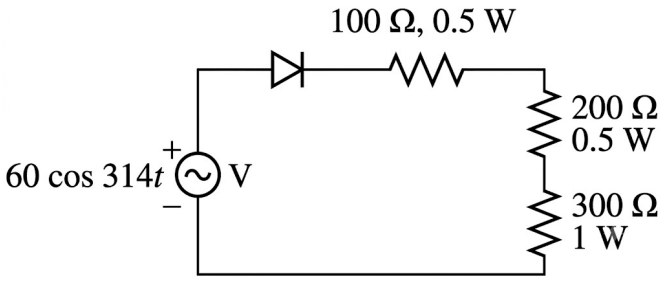


Fig. Q1.04.1

A. 0.25 W **B.** 0.50 W **C.** 0.75 W **D.** 1.00 W

Q1.01.1 MCQ 2001 1M

In the circuit shown in below figure, current through the 5 Ω resistor is

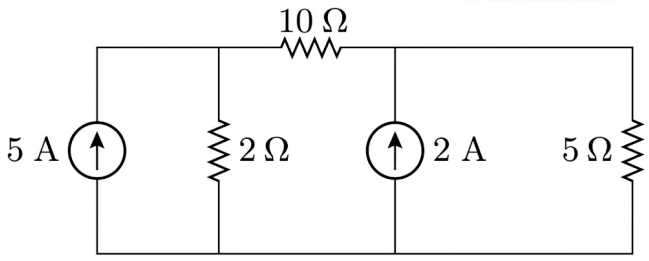


Fig. Q1.01.1

A. 0 A **B.** 2 A **C.** 3 A **D.** 7 A

Q1.99.1 MCQ 1999 1M

For the waveform $v(t) = 2+\cos\left(\omega t + \frac{\pi}{6}\right)$, the ratio $\frac{V_{rms}}{V_{average}}$ is

A. $\frac{3}{2\sqrt{2}}$ **B.** $\sqrt{\frac{3}{2}}$

C. π **D.** $\frac{\pi}{2}$

Q1.99.2 MCQ 1999 1M

The bridge circuit in given figure is balanced. The magnitude of current I is

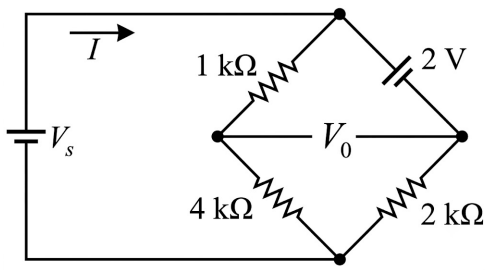


Fig. Q1.99.2

A. 2 mA **B.** 4 mA **C.** 5 mA **D.** 6 mA

Q1.96.1 MCQ 1996 1M

A voltage drop across a non-linear element is given by $v = i + 2i^2$. The power loss across it for a current of sin t is

A. 1.0 W **B.** 2.0 W **C.** 3.0 W **D.** 0.5 W

Answer Key

Q1.26.1 B	Q1.25.1 C	Q1.23.1 1
Q1.20.1 A	Q1.20.2 C	Q1.19.1 1000
Q1.18.1 A	Q1.17.1 2	Q1.16.1 10
Q1.15.1 1	Q1.15.2 1.70 to 1.80	Q1.13.1 B
Q1.13.2 C	Q1.12.1 A	Q1.11.1 A
Q1.10.1 C	Q1.09.1 C	Q1.08.1 D
Q1.08.2 D	Q1.08.3 A	Q1.08.4 D
Q1.06.1 B	Q1.06.2 C	Q1.04.1 C
Q1.01.1 B	Q1.99.1 A	Q1.99.2 D
Q1.96.1 D		

UNIT

II

Transient Analysis

Topics

1. Introduction to Transient Analysis
2. Basics of Signals for Transient Analysis
3. RC Circuits — Introduction and Examples
4. Impulse Current in Capacitor and Basics of Laplace in Circuit Analysis
5. Slope at $t = 0^+$, Switched Capacitor and Impulse Response of RC Circuits
6. RC Circuits with Pulse Wave Input
7. Initial and Final Value Theorems
8. RL Circuits

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Unit Snapshot*

Stream: IN	**Questions:** 24	**MCQ:** 11	**MSQ:** 0	**NAT:** 13
1M: 7	**2M:** 17	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2025)–4M(2017)

**See preface for how Avg Weightage and Range are calculated.*

Q2.25.1 NAT 2025 1M

In the circuit shown, the switch is opened at $t = 0$ s. The current $i(t)$ at $t = 2$ ms is _____ mA (rounded off to two decimal places).

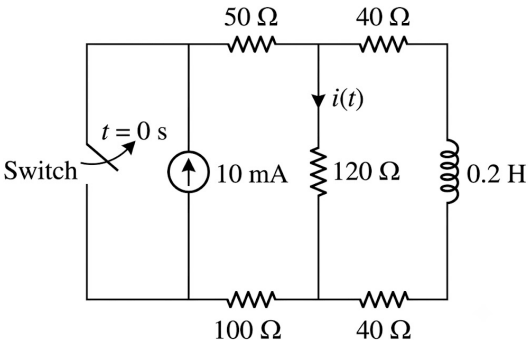


Fig. Q2.25.1

Q2.24.1 NAT 2024 2M

The switch in the following figure has been closed for a long time ($t < 0$). It is opened at $t = 0$ seconds. The value of $\frac{dv_c}{dt}$ at $t = 0^+$ is _____ V/s (rounded off to nearest integer).

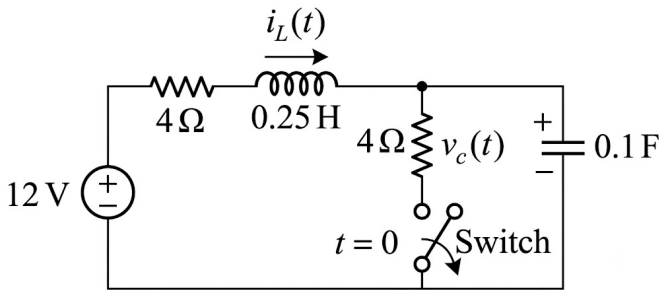


Fig. Q2.24.1

Q2.23.1 NAT 2023 2M

The RL circuit with $R = 10$ kΩ and $L = 1$ mH is excited by a step current $I_0u(t)$. At $t = 0^-$, there is a current $I_L = I_0/5$ flowing through the inductor. The minimum time taken for the current through the inductor to reach 99% of its final value is _____ μs (rounded off to two decimal places).

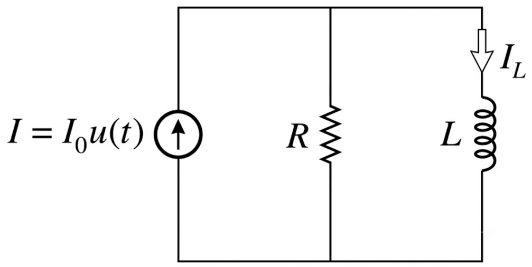


Fig. Q2.23.1

Q2.22.1 NAT 2022 2M

In the circuit shown, the capacitance $C_0 = 10$ μF and inductance $L_0 = 1$ mH and the diode is ideal. The capacitor is initially charged to 10 V and the current in the inductor is initially zero. If the switch is closed at $t = 0$ sec, the voltage $V_c(t)$ (in volts) across the capacitor at $t = 0.5$ sec is _____ (round off to one decimal place).

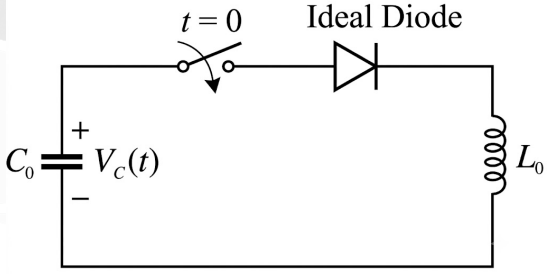


Fig. Q2.22.1

Q2.19.1 NAT 2019 1M

In the circuit shown below, initially the switch S_1 is open, the capacitor C_1 has a charge of 6 coulomb, and the capacitor C_2 has 0 coulomb. After S_1 is closed, the charge on C_2 in steady state is _____ coulomb.

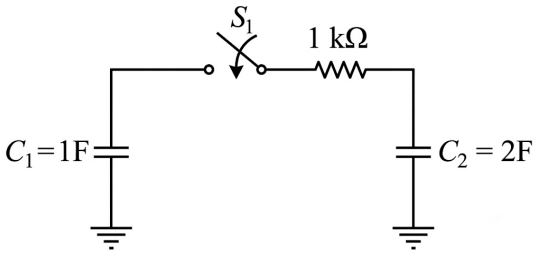


Fig. Q2.19.1

Q2.19.2 NAT 2019 2M

In the circuit shown below, a step input voltage of magnitude 5 V is applied at node A at time $t = 0$. If the capacitor has no charge for $t \leq 0$, the voltage at node P at $t = 6\ \mu s$ is _____ V. (Answer should be rounded off to two decimal places.)

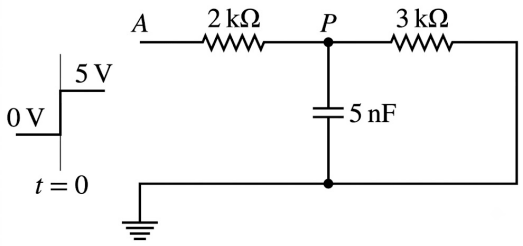


Fig. Q2.19.2

Q2.17.1 NAT 2017 2M

In the circuit diagram shown in the figure, S_1 was closed and S_2 was open for a very long time. At $t = 0$, S_1 is opened and S_2 is closed. The voltage across the capacitor, in volt, at $t = 5\ \mu s$ is _____.

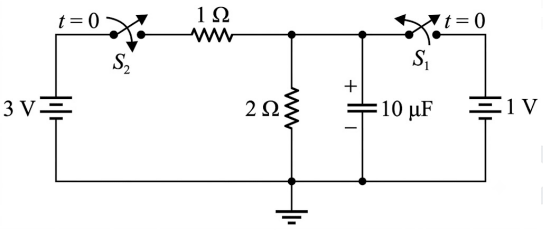


Fig. Q2.17.1

Q2.17.2 NAT 2017 2M

The current response of a series RL circuit to a unit step voltage is given in the table. The value of L is _____ H.

t (s)	0	0.25	0.5	0.75	1.0	∞
$i(t)$ (A)	0	0.197	0.316	0.388	0.432	0.5

Q2.16.1 NAT 2016 2M

The voltage $v(t)$ shown below is applied to the given circuit. $v(t) = 3$ V for $t < 0$ and $v(t) = 6$ V for $t > 0$. The value of current $i(t)$ at $t = 1$ sec, in ampere, is _____.

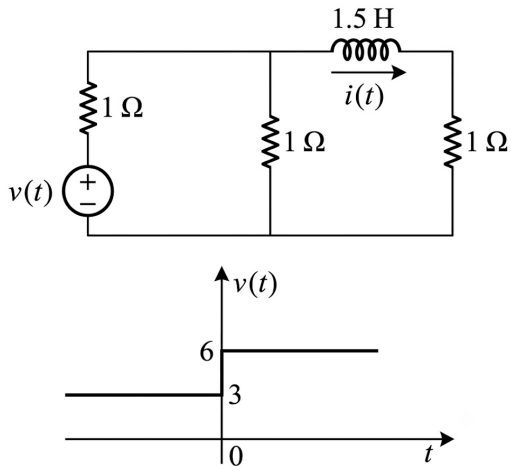


Fig. Q2.16.1

Q2.16.2 NAT 2016 1M

A current $i(t)$ shown in the figure below is passed through a 1 F capacitor that had zero initial charge. The voltage across the capacitor for $t > 2$ s in volt is _____.

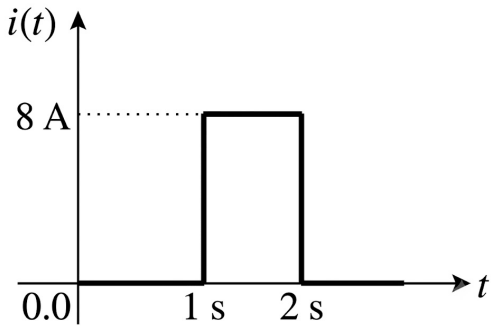


Fig. Q2.16.2

Q2.15.1 NAT 2015 1M

The capacitor shown in the figure is initially charged to 10 V. The switch closes at time $t = 0$. Then the value of $V_c(t)$ in volts at time $t = 10$ ms is _____ V.

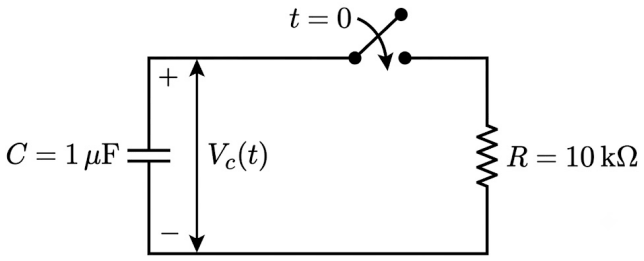


Fig. Q2.15.1

Q2.14.1 NAT 2014 2M

In the circuit shown in the figure, initially the capacitor is uncharged. The switch S is closed at $t = 0$. Two milliseconds after the switch is closed, the current through the capacitor (in mA) is _____.

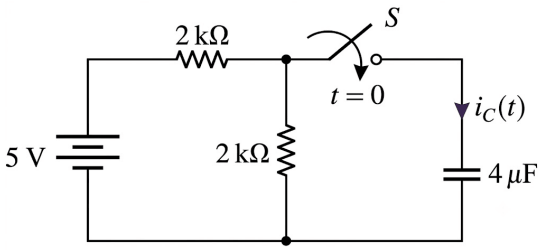


Fig. Q2.14.1

Q2.14.2 NAT 2014 1M

The circuit shown in the figure was at steady state for $t < 0$ with the switch at position A. The switch is thrown to position B at time $t = 0$. The voltage V (volts) across the 10 Ω resistor at time $t = 0^+$ is _____.

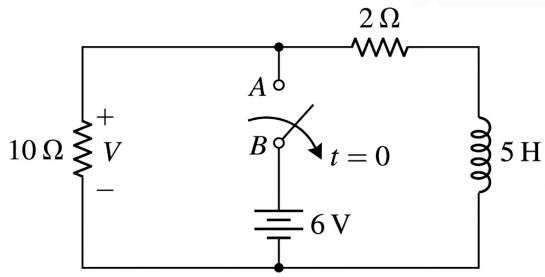


Fig. Q2.14.2

Q2.11.1 MCQ 2011 1M

In the circuit shown below, the switch, initially at position 1 for a long time, is changed to position 2 at $t = 0$. The current i through the inductor for $t \geq 0$ is

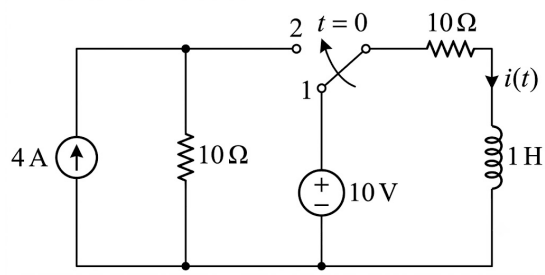


Fig. Q2.11.1

A. $1 - e^{-20t}$ A **B.** $1 + e^{-20t}$ A
C. $1 + 2e^{-20t}$ A **D.** $2 - e^{-20t}$ A

Q2.10.1 MCQ 2010 1M

In the dc circuit shown in the adjoining figure, the node voltage V_2 at steady state is

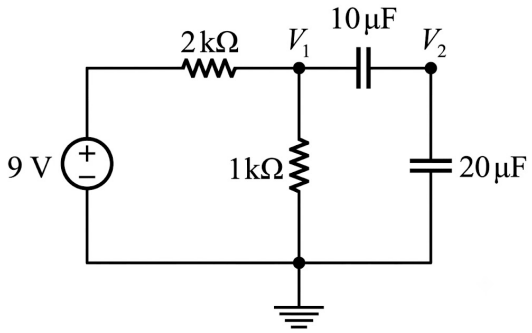


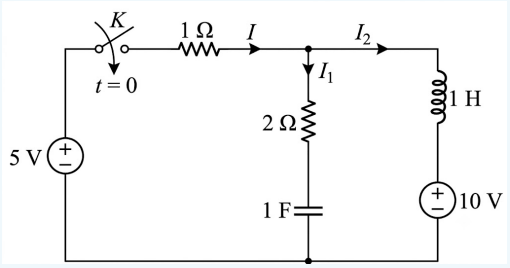
Fig. Q2.10.1

A. 0 V **B.** 1 V **C.** 2 V **D.** 3 V

Common Data Questions

The next 2 questions share the data below.

In the circuit shown below, the steady state is reached with the switch K open. Subsequently, the switch is closed at time $t = 0$.



Q2.08.1 MCQ 2008 2M

At time $t = 0^+$, current I is

A. $-\frac{5}{3}$ A **B.** 0 A **C.** $\frac{5}{3}$ A **D.** ∞

Q2.08.2 MCQ 2008 2M

At time $t = 0^+$, $\frac{dI_2}{dt}$ is

A. −5 A/s **B.** $-\frac{10}{3}$ A/s **C.** 0 A/s **D.** 5 A/s

Q2.07.1 MCQ 2007 2M

In the circuit shown in the following figure, the switch is kept closed for a long time and then opened at $t = 0$. The value of the current i just before opening the switch ($t = 0^-$) and just after opening the switch ($t = 0^+$) are, respectively,

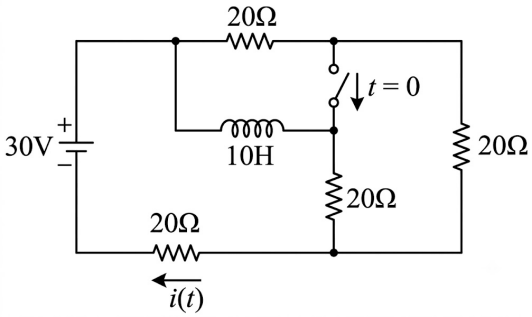


Fig. Q2.07.1

A. $\frac{3}{4}$ A and 1 A

B. $\frac{7}{6}$ A and $\frac{5}{2}$ A

C. 1 A and $\frac{7}{6}$ A

D. 1 A and 1 A

Q2.06.1 MCQ 2006 2M

In the circuit shown in the following figure, the input voltage $v_i(t)$ is constant at 2 V for time $t < 1$ s and then it changes to 1 V. The output voltage, $v_o(t)$, 2 s after the change will be

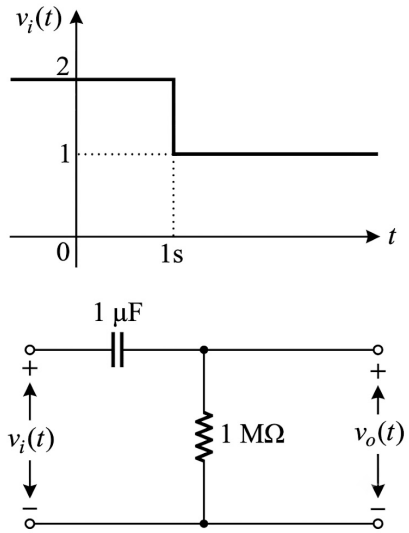


Fig. Q2.06.1

A. $-\exp(-2)$ Volt

B. $-1 + \exp(-2)$ Volt

C. $\exp(-2)$ Volt

D. $1 - \exp(-2)$ Volt

Q2.05.1 MCQ 2005 2M

In the circuit shown, the switch S is closed at time $t = 0$. The supply current at $t = 0^+$ and capacitor voltage at $t \to \infty$ are, respectively.

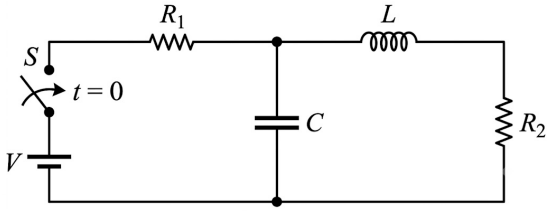


Fig. Q2.05.1

A. 0 and V

B. $\frac{V}{R_1}$ and $\frac{VR_2}{R_1 + R_2}$

C. $\frac{V}{R_1 + R_2}$ and $\frac{VR_2}{R_1 + R_2}$

D. $\frac{V}{R_1}$ and V

Q2.00.1 MCQ 2000 2M

In the circuit shown, $C_1 = C_2 = 2$ F and $L = 1$ H. Initially C_1 and C_2 are charged to 20 V and 0 V, respectively. If the switch is closed at $t = 0$, then $V_{C_1}(t)$ and $V_{C_2}(t)$ are (use the Laplace transform approach).

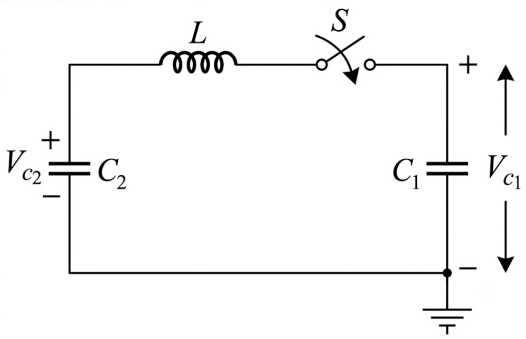


Fig. Q2.00.1

A. $V_{C_1}(t) = 10(1 + \cos t)u(t)$ and $V_{C_2}(t) = 10(1 - \cos t)u(t)$

B. $V_{C_1}(t) = 10(1 + \cos t)u(t)$ and $V_{C_2}(t) = 10(1 + \cos t)u(t)$

C. $V_{C_1}(t) = 10(1 - \cos t)u(t)$ and $V_{C_2}(t) = 10(1 - \cos t)u(t)$

D. $V_{C_1}(t) = 10(1 - \cos t)u(t)$ and $V_{C_2}(t) = 10(1 + \cos t)u(t)$

Q2.99.1 MCQ 1999 2M

The output voltage of the circuit shown in figure for $t > 0$ will be

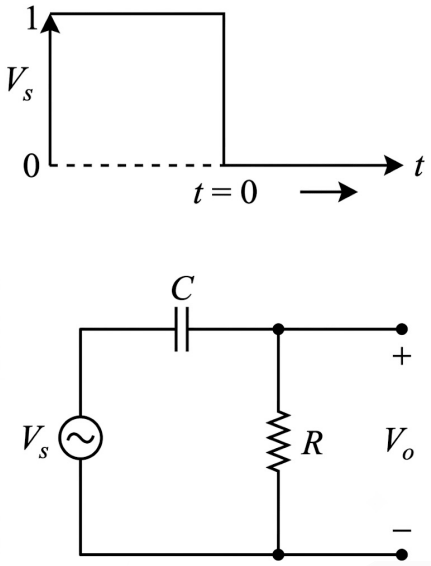


Fig. Q2.99.1

A. $e^{-t/RC}$ **B.** $-e^{-t/RC}$ **C.** $1 - e^{-t/RC}$ **D.** $e^{-t/RC} - 1$

Q2.98.1 MCQ 1998 2M

In the given figure, the stored energy of the capacitor is

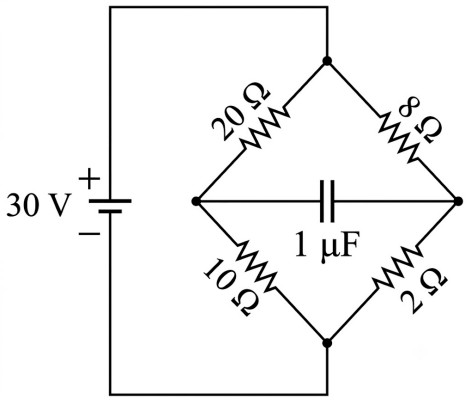


Fig. Q2.98.1

A. 8 μJ **B.** 30 μJ **C.** 450 μJ **D.** 900 μJ

Q2.95.1 MCQ 1995 2M

The circuit shown in the figure is supplied by a DC source. The switch (S) is closed at $t = 0$ and remains closed for a long time. The steady-state output voltage (V_0) is

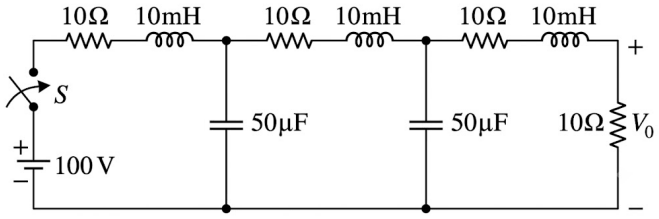


Fig. Q2.95.1

A. 40 V **B.** 100 V **C.** 25 V **D.** 20 sin t V

Answer Key

Q2.25.1 4.60 to 5.00	Q2.24.1 15	Q2.23.1 0.43 to 0.45
Q2.22.1 -10.1 to -9.9	Q2.19.1 4	Q2.19.2 1.87 to 1.91
Q2.17.1 1.4 to 1.6	Q2.17.2 0.9 to 1.1	Q2.16.1 1.5 to 1.7
Q2.16.2 8	Q2.15.1 3.6 to 3.7	Q2.14.1 1.5 to 1.6
Q2.14.2 -31 to -29	Q2.11.1 D	Q2.10.1 B
Q2.08.1 A	Q2.08.2 B	Q2.07.1 C
Q2.06.1 A	Q2.05.1 B	Q2.00.1 A
Q2.99.1 B	Q2.98.1 A	Q2.95.1 C

UNIT

III

Network Theorems For DC & Miscellaneous Topics

Topics

1. Source Transformation and Superposition Theorem
2. Thevenin's Theorem
3. Norton's Theorem
4. Concept of Linearity
5. Reciprocity Theorem
6. Maximum Power Transfer Theorem
7. Star-Delta Connections and Conversions
8. Finding Equivalent Resistances
9. Equivalent Resistance of a Cube Across Different Nodes
10. Equivalent Resistance in an Infinite Ladder
11. Bulb-Based Problems

Unit Snapshot*

Stream: IN	**Questions:** 21	**MCQ:** 14	**MSQ:** 0	**NAT:** 7
1M: 10	**2M:** 11	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2023)–3M(2018)

**See preface for how Avg Weightage and Range are calculated.*

Q3.25.1 NAT 2025 2M

In the circuit shown, the galvanometer (G) has an internal resistance of 100 Ω. The galvanometer current I_G is _____ μA (rounded off to the nearest integer).

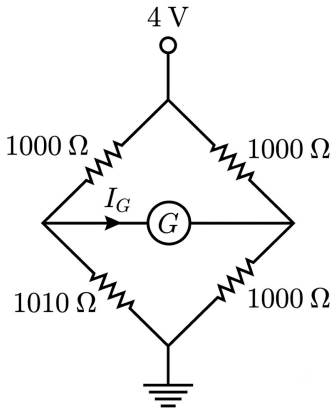


Fig. Q3.25.1

Q3.23.1 NAT 2023 1M

In the circuit shown, the Thevenin equivalent resistance R_{th} across the terminals a and b is _____ Ω (rounded off to one decimal place).

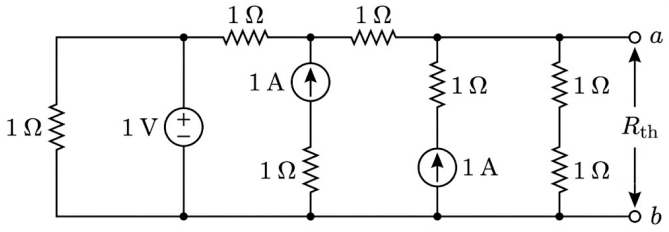


Fig. Q3.23.1

Q3.22.1 NAT 2022 2M

Given circuit A with currents I_1 and I_2 as shown, the current I_3 in circuit B (in amperes), is _____. (round off to one decimal place)

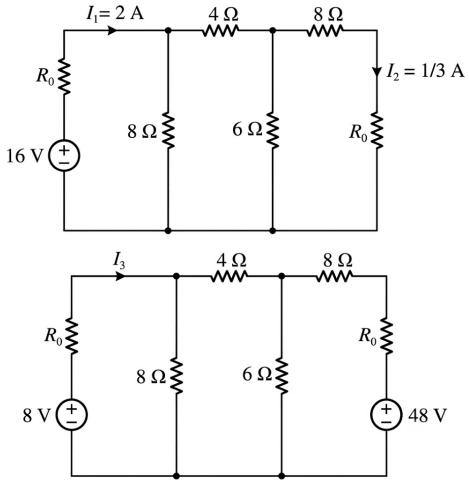


Fig. Q3.22.1

Q3.19.1 NAT 2019 1M

In the circuit shown below, maximum power is transferred to the load resistance R_L, when R_L = _____ Ω.

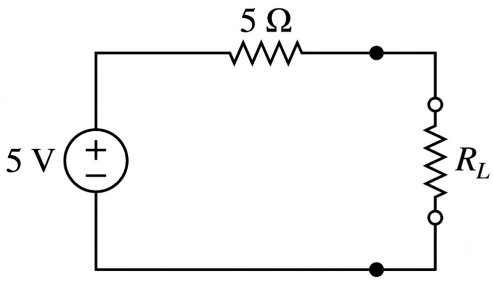


Fig. Q3.19.1

Q3.18.1 MCQ 2018 1M

The Thevenin equivalent circuit representation across terminals p–q of the circuit shown in the figure is

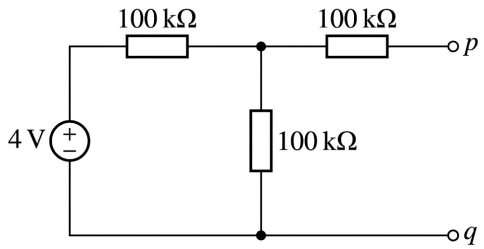


Fig. Q3.18.1

A. 1 V source in series with 150 kΩ

B. 1 V source in parallel with 100 kΩ

C. 2 V source in series with 150 kΩ

D. 2 V source in parallel with 200 kΩ

Q3.18.2 NAT 2018 2M

In the given circuit, superposition is applied. When V_2 is set to 0 V, the current I_2 is −6 A. When V_1 is set to 0 V, the current I_1 is +6 A. Current I_3 (in A) when both sources are applied will be (up to two decimal places) _____.

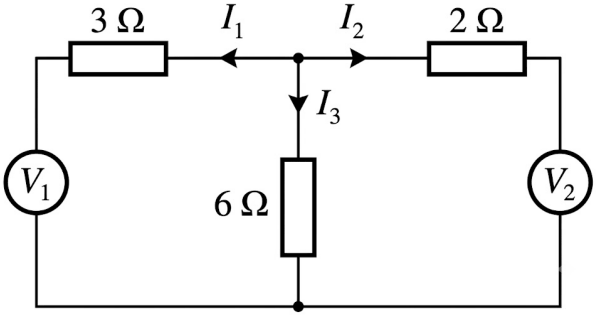


Fig. Q3.18.2

Q3.15.1 MCQ 2015 1M

A load resistor R_L is connected to a battery of voltage E with internal resistance R_i through a resistance R_s as shown in the figure. For fixed values of R_L and R_i, the value of R_s (≥ 0) for maximum power transfer to R_L is

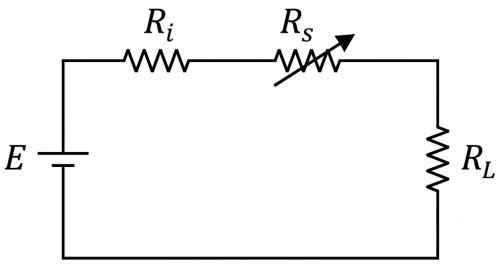


Fig. Q3.15.1

A. 0 **B.** $R_L - R_i$ **C.** R_L **D.** $R_L + R_i$

Q3.15.2 NAT 2015 2M

Consider the circuits shown in the figure. The magnitude of the ratio of the currents, i.e. $\left|\frac{I_1}{I_2}\right|$, is _____.

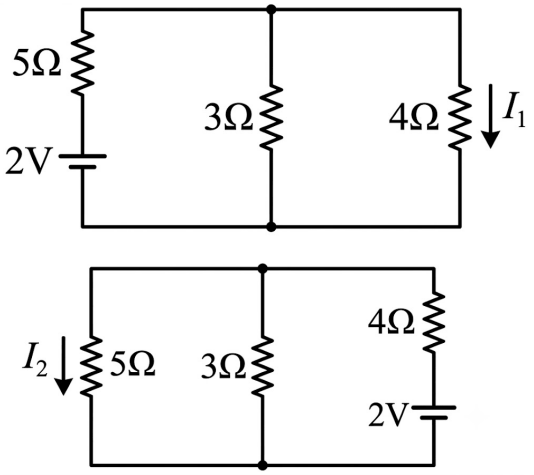


Fig. Q3.15.2

Q3.14.1 NAT 2014 2M

The circuit shown in the figure contains a dependent current source between A and B terminals. The Thevenin's equivalent resistance (in kΩ) between terminals C and D is _____.

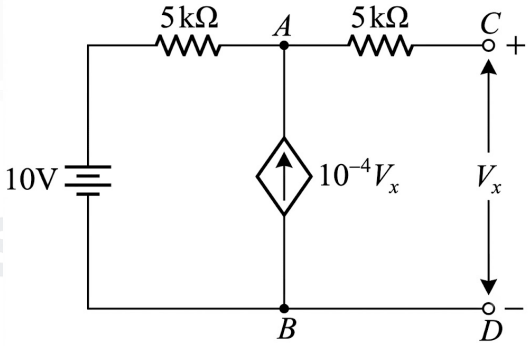


Fig. Q3.14.1

Q3.13.1 MCQ 2013 1M

A source $v_s(t) = V\cos 100\pi t$ volt has an internal impedance of $(4 + j3)\,\Omega$. If a purely resistive load connected to this source has to extract the maximum power out of the source, its value in Ω should be

A. 3 **B.** 4 **C.** 5 **D.** 7

Q3.13.2 MCQ 2013 2M

In the circuit shown below, if the source voltage $V_s = 100\angle 53.13°$ V, then the Thevenin's equivalent voltage (in volts) as seen by the load resistance R_L is

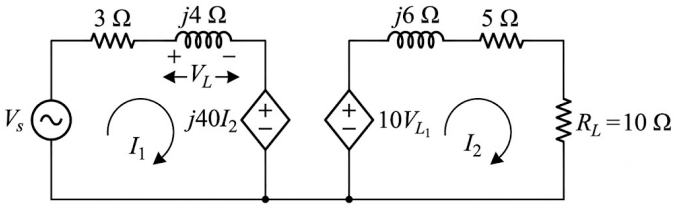


Fig. Q3.13.2

A. $100\angle 90°$ **B.** $800\angle 0°$
C. $800\angle 90°$ **D.** $100\angle 60°$

Q3.12.1 MCQ 2012 1M

The impedance looking into nodes 1 and 2 in the given circuit is

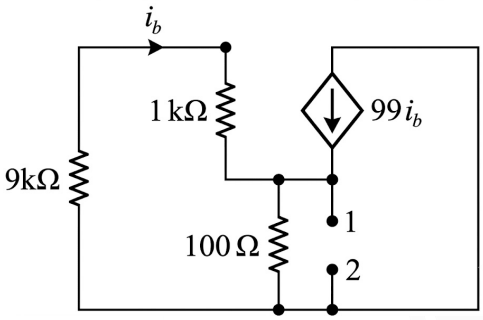


Fig. Q3.12.1

A. 50 Ω **B.** 100 Ω **C.** 5 kΩ **D.** 10.1 kΩ

Q3.12.2 MCQ 2012 2M

Assuming both the voltage sources are in phase, the value of R for which maximum power is transferred from circuit A to circuit B is

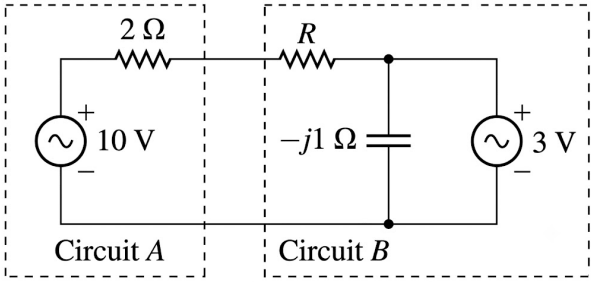


Fig. Q3.12.2

A. 0.8 Ω **B.** 1.4 Ω **C.** 2 Ω **D.** 2.8 Ω

Q3.09.1 MCQ 2009 2M

The source network S is connected to the load network L as shown by dashed lines. The power transferred from S to L would be maximum when R_L is

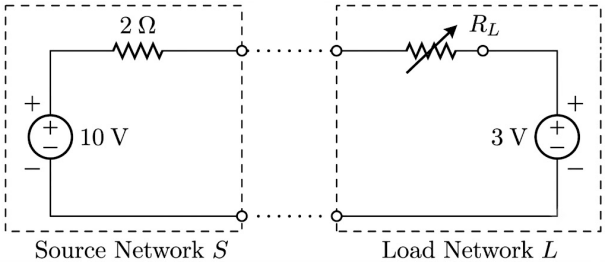


Fig. Q3.09.1

A. 0 Ω **B.** 0.6 Ω **C.** 0.8 Ω **D.** 2.0 Ω

Q3.08.1 MCQ 2008 2M

In the circuit shown below, the maximum power that can be transferred to the load Z_L is

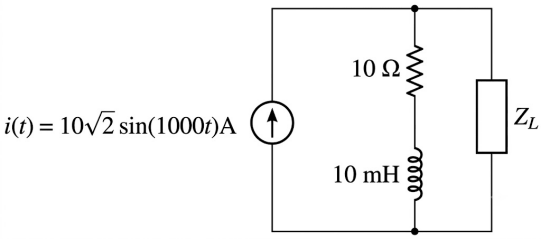


Fig. Q3.08.1

A. 250 W **B.** 500 W **C.** 1000 W **D.** 2000 W

Q3.07.1 MCQ 2007 1M

In full sunlight, a solar cell has a short circuit current of 75 mA and a current of 70 mA for a terminal voltage of 0.6 V with a given load. The Thevenin resistance of the solar cell is

A. 8 Ω **B.** 8.6 Ω **C.** 120 Ω **D.** 240 Ω

Q3.07.2 MCQ 2007 2M

In the circuit shown in the following figure, the current through the 1 Ω resistor is

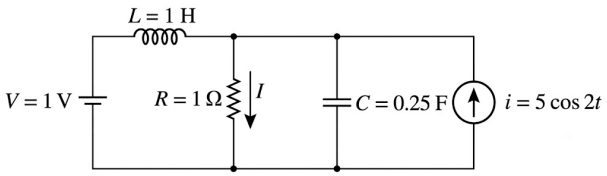


Fig. Q3.07.2

A. $(1 + 5\cos 2t)$ A **B.** $(5 + \cos 2t)$ A
C. $(1 - 5\cos 2t)$ A **D.** 6 A

Q3.03.1 MCQ 2003 1M

The output resistance across terminals 1 and 2 of the DC bridge shown in the figure is

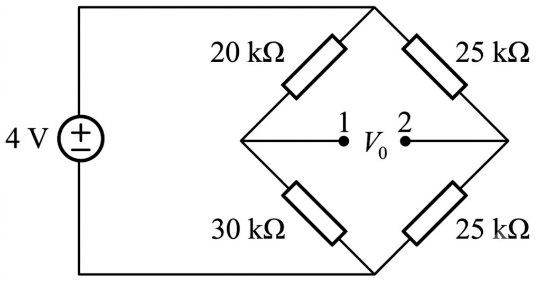


Fig. Q3.03.1

A. 12.5 kΩ **B.** 24.5 kΩ **C.** 25.0 kΩ **D.** 100 kΩ

Q3.00.1 MCQ 2000 1M

For the circuit shown in figure, $V_R = 20$ V when $R = 10\ \Omega$ and $V_R = 30$ V when $R = 20\ \Omega$. For $R = 80\ \Omega$, V_R will be

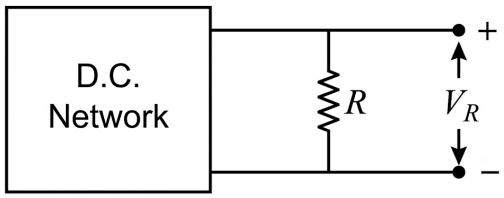


Fig. Q3.00.1

A. 48 V **B.** 60 V **C.** 120 V **D.** 160 V

Q3.00.2 MCQ 2000 1M

For the circuit shown in figure, R is adjusted to have maximum power transferred to it. The maximum power transferred is

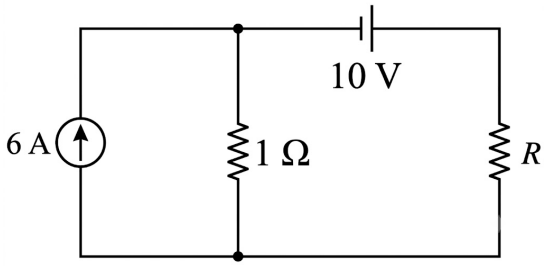


Fig. Q3.00.2

A. 16 W **B.** 32 W **C.** 64 W **D.** 100 W

Q3.00.3 MCQ 2000 2M

Superposition theorem does not hold for

A. R-L-C networks

B. networks containing diodes and R-C element

C. linear op-amp circuits

D. none

Answer Key

Q3.25.1 9	Q3.23.1 1	Q3.22.1 0
Q3.19.1 5	Q3.18.1 C	Q3.18.2 0.95 to 1.05
Q3.15.1 A	Q3.15.2 1	Q3.14.1 20
Q3.13.1 C	Q3.13.2 C	Q3.12.1 A
Q3.12.2 A	Q3.09.1 C	Q3.08.1 B
Q3.07.1 C	Q3.07.2 A	Q3.03.1 B
Q3.00.1 A	Q3.00.2 C	Q3.00.3 B

UNIT

IV

2nd Order Dynamics and Complex Impedance

Topics

1. Source-Free 2nd Order RLC Circuits
2. Initial Conditions and Sourced RLC Circuits
3. LC Oscillators

PrepFusion

Unit Snapshot*

Stream: IN	**Questions:** 2	**MCQ:** 1	**MSQ:** 0	**NAT:** 1
1M: 0	**2M:** 1	**Descriptive:** 1	**Avg Weightage** ~0M	**Range:** 0M(–)–0M(–)

**See preface for how Avg Weightage and Range are calculated.*

Q4.01.1 NAT 2001 5M

In the circuit shown in Fig. 14, $C_1 = C_2 = 2\,\text{F}$ and $\text{L} = 1\,\text{H}$. Initially C_1 and C_2 are charged to voltages 20 and 0 volts, respectively. If the switch is closed at t = 0, determine $v_{c1}(t)$ and $v_{c2}(t)$. Use the Laplace transform approach.

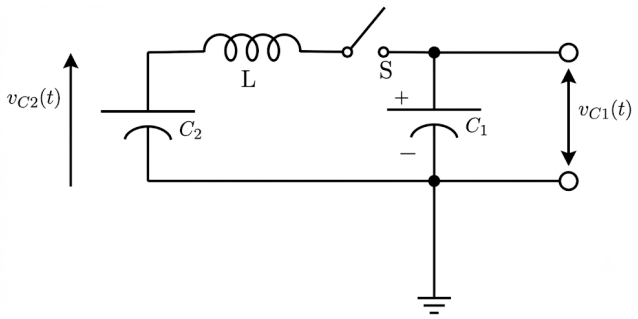


Fig. Q4.01.1

Q4.00.1 MCQ 2000 2M

The value of the impedance $Z_d(s)$ between 1-1 of the circuit shown in Fig 1.27 as $s \to \infty$ is

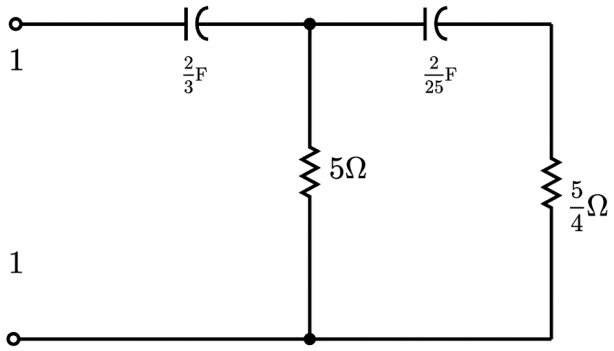


Fig. Q4.00.1

A. 0 **B.** 1 **C.** 5/4 **D.** 5

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q4.01.1	Q4.00.1	
*	B	

UNIT

V

AC Circuit Analysis

Topics

1. Basics of Phasor Diagrams
2. Phasor Diagram for Series RL, RC and RLC Circuits
3. Phasor Diagram for Parallel RL, RC and RLC Circuits
4. Locus Diagram of Current, Admittance and Impedance
5. Introduction to Power in AC Circuits
6. Active, Reactive and Apparent Power in Complex Networks
7. Series RLC Resonance Circuit
8. Quality Factor and Stored Energy in Series Resonance
9. Parallel RLC Resonance Circuit
10. Application of Network Theorems in AC Circuits
11. Maximum Power Transfer in AC Circuits
12. Three Phase Circuits

Unit Snapshot*

Stream: IN	**Questions:** 57	**MCQ:** 29	**MSQ:** 0	**NAT:** 28
1M: 23	**2M:** 34	**Descriptive:** 0	**Avg Weightage** ~5M	**Range:** 2M(2026)–8M(2012)

**See preface for how Avg Weightage and Range are calculated.*

Q5.26.1 MCQ 2026 1M

The RLC filter shown below acts as a _____.

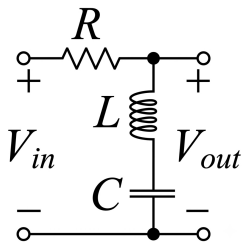


Fig. Q5.26.1

A. low pass filter
B. high pass filter
C. band pass filter
D. band stop filter

Q5.26.2 NAT 2026 1M

In a series RL circuit shown in the figure, the current through the resistor lags the AC source voltage by 45°. Let $R = 100\pi\ \Omega$ and L = 2 H.

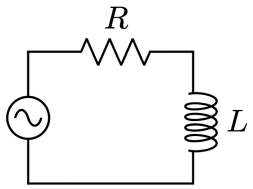


Fig. Q5.26.2

The frequency of the source is _____ Hz. *(rounded off to the nearest integer)*

Q5.25.1 MCQ 2025 1M

Three parallel admittances $Y_a = -j0.2S$, $Y_b = 0.3S$ and $Y_c = j0.4S$ connected in parallel with a voltage source $V_s = 10\angle 45°$ V draws a total current I_s from the source. The currents flowing through each of these admittances are I_a, I_b and I_c respectively. Let $I = I_b + I_c$. The phase relation between I and I_s is

A. I leads I_s by 19.44°

B. I lags I_s by 19.44°

C. I leads I_s by 33.69°

D. I lags I_s by 33.69°

Q5.25.2 NAT 2025 2M

For the circuit shown in the figure, the active power supplied by the source is _____ W **(rounded off to one decimal place)**.

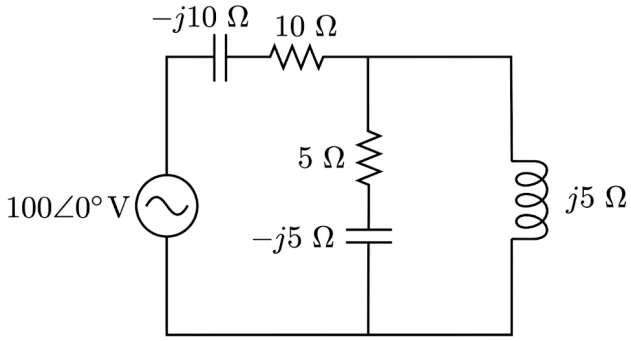


Fig. Q5.25.2

Q5.24.1 NAT 2024 1M

The voltage applied and the current drawn by a circuit are
$v(t) = 95 + 200\cos(120\pi t) + 90\cos(360\pi t - 60°)$ V
$i(t) = 4\cos(120\pi t - 60°) + 1.5\cos(240\pi t - 75°)$ A
The average power absorbed by the circuit is _____ W **(rounded off to nearest integer).**

Q5.24.2 NAT 2024 1M

The current i(t) drawn by a circuit is given as
$i(t) = 4 + 30\cos(t) - 20\sin(t) + 15\cos(3t) - 10\sin(3t)$ A
The root-mean-square value of i(t) is _____ A **(rounded off to one decimal place).**

Q5.23.1 NAT 2023 2M

In the circuit shown, $\omega = 100\pi$ rad/s, $R_1 = R_2 = 2.2\Omega$ and L=7mH.The capacitance C for which Y_{in} is purely real is _____ mF **(rounded off to two decimal places).**

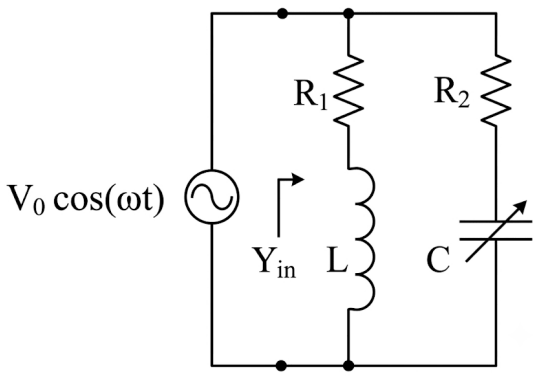


Fig. Q5.23.1

Q5.23.2 NAT 2023 2M

The voltage source $V_S = 10\sqrt{2}\sin(20000\pi t)V$ has an internal resistance of 50Ω. The RMS value of the current through R is _____ mA **(rounded off to one decimal place).**

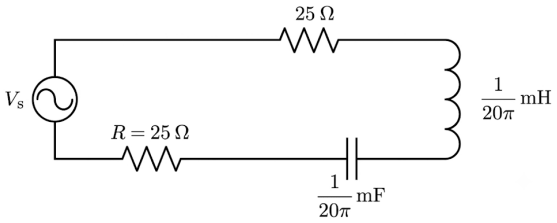


Fig. Q5.23.2

Q5.22.1 NAT 2022 1M

In the bandpass filter circuit shown, $R_0 = 50\Omega, L_0 = 1$ mH, $C_0 = 10$ nF. The Q factor of the filter is _____ (round off to two decimal places)

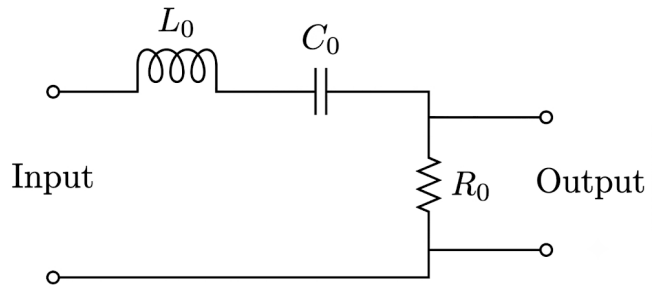


Fig. Q5.22.1

Q5.22.2 NAT 2022 2M

In the circuit shown, the load is driven by a sinusoidal ac voltage source $V_1 = 100\angle 0°$ V at 50 Hz. Given $R_1 = 20\ \Omega$, $C_1 = \left(\frac{1000}{\pi}\right)\mu$F, $L_1 = \left(\frac{20}{\pi}\right)$ mH and $R_2 = 4\ \Omega$, the power factor is _____ (round off to one decimal place)

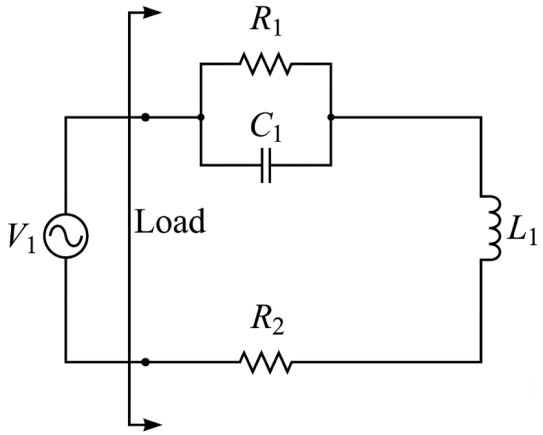


Fig. Q5.22.2

Q5.21.1 NAT 2021 2M

A household fan consumes 60 W and draws a current of 0.3125 A (rms) when connected to a 230 V (rms) ac, 50 Hz single phase mains. The reactive power drawn by the fan in VAR is _____ (rounded off to the nearest integer).

Q5.21.2 NAT 2021 2M

An air cored coil having a winding resistance of 10Ω is connected in series with a variable capacitor C_x. The series circuit is excited by a 10 V sinusoidal voltage source of angular frequency 1000 rad/s. As the value of the capacitor is varied, a maximum voltage of 30V was observed across it. Neglecting skin-effect, the value of the inductance of the coil in millihenry is _____.

Q5.20.1 NAT 2020 1M

The capacitance C_x of a capacitive type sensor is (1000 x) pF, where x is the input to the sensor. As shown in the figure, the sensor is excited by a voltage $10\sin(100\pi t)$ V The other terminal of the sensor is tied to the input of a high input impedance amplifier through a shielded cable, with shield connected to ground. The cable capacitance is 100 pF. The peak of the voltage V_A at the input of the amplifier when x = 0.1 (in volts) is _____

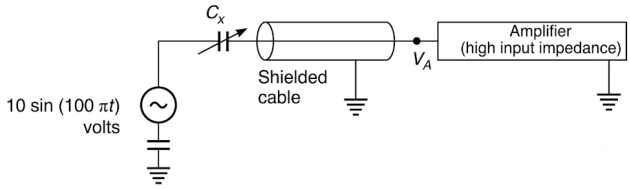


Fig. Q5.20.1

Q5.20.2 NAT 2020 2M

The rms value of the phasor current $\underline{I}$ in the circuit shown **(in amperes)** is _____

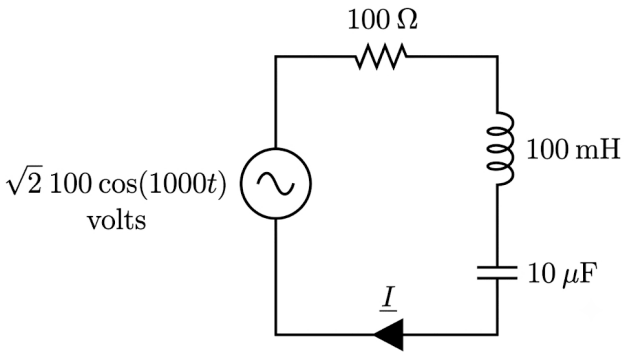


Fig. Q5.20.2

Q5.20.3 NAT 2020 1M

Three 400Ω resistors are connected in delta and powered by a 400 V (rms), 50 Hz, balanced, symmetrical R-Y-B sequence, three-phase three- wire mains. The rms value of the **line current (in amperes, rounded off to one decimal place)** is _____

Q5.20.4 NAT 2020 2M

In the circuit shown, the rms value of the voltage across the 100Ω resistor **(in volts)** is _____

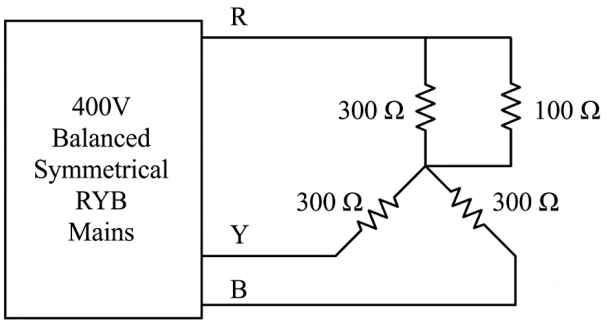


Fig. Q5.20.4

Q5.19.1 MCQ 2019 1M

If each of the values of inductance, capacitance and resistance of a series LCR circuit are doubled, the Q factor of the circuit would

A. reduce by a factor of $\sqrt{2}$

B. reduce by a factor of 2

C. increase by a factor of $\sqrt{2}$

D. increase by a factor of 2

Q5.19.2 NAT 2019 2M

In the circuit shown below, the angular frequency ω at which the current is in phase with the voltage is _____ rad/s.

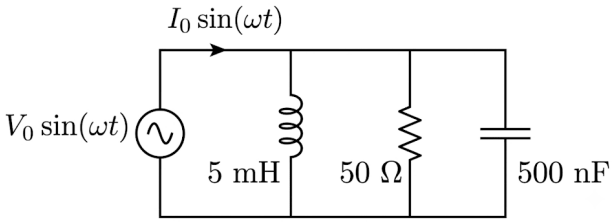
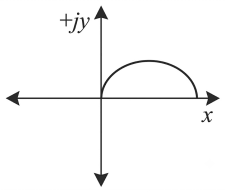
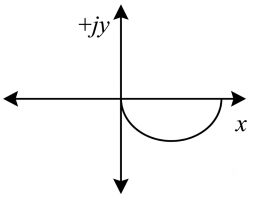


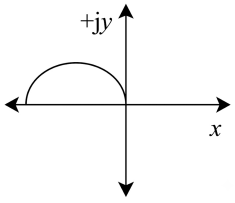
Fig. Q5.19.2

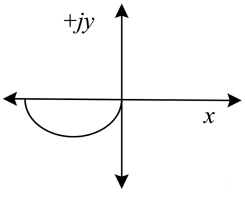
Q5.18.1 MCQ 2018 1M

A series R-C circuit is excited by a 1∠0 V sinusoidal ac voltage source. The locus diagram of the phasor current $\underline{I}$ = (x+jy) A, when C is varied, while keeping R fixed, is

A.


B.


C.


D.


Q5.18.2 NAT 2018 1M

A coil having an impedance of (10+j100) Ω is connected in parallel to a variable capacitor as shown in figure. Keeping the excitation frequency unchanged, the value of the capacitor is changed so that parallel resonance occurs. The impedance across terminals p-q at resonance (in Ω) is _____.

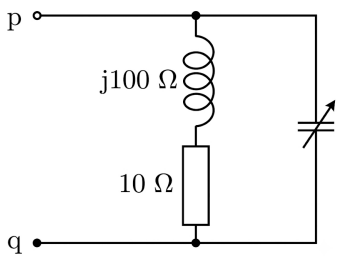


Fig. Q5.18.2

Q5.18.3 NAT 2018 2M

In the figure, an RLC load is supplied by a 230 V, 50 Hz single phase source. The magnitude of the reactive power (in VAR) supplied by the source is _____

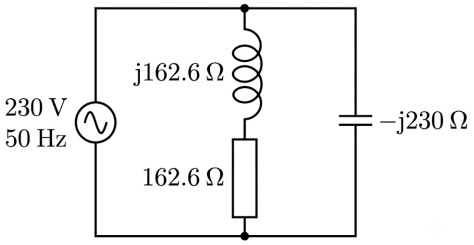


Fig. Q5.18.3

Q5.17.1 NAT 2017 1M

A series R-L-C circuit is excited by a 50 V, 50 Hz sinusoidal source. The voltages across the resistance and the capacitance are shown in the figure. The voltage across the inductor (V_L) is _____.

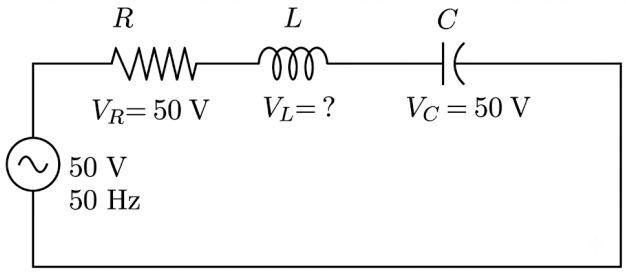


Fig. Q5.17.1

Q5.17.2 NAT 2017 2M

For the circuit shown in the figure, the total real power delivered by the source to the loads is _____ kW.

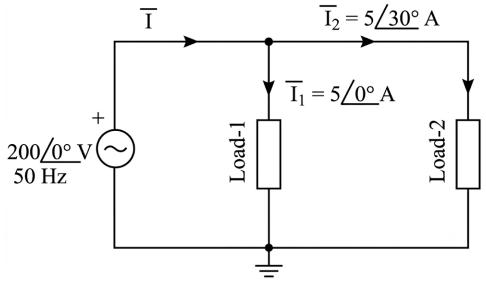


Fig. Q5.17.2

Q5.17.3 NAT 2017 2M

A series R-L-C circuit is excited with an a.c. voltage source. The quality factor (Q) of the circuit is given as Q=30.The amplitude of the current in **ampere** at upper half - power frequency will be _____.

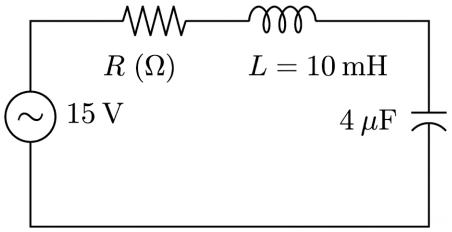


Fig. Q5.17.3

Q5.16.1 NAT 2016 1M

Three currents i_1, i_2 and i_3 meet at a node as shown in the figure below. If $i_1 = 3\cos(\omega t)$ ampere, $i_2 = 4\sin(\omega t)$ ampere and $i_3 = I_3\cos(\omega t + \theta)$ ampere, the value of I_3 in **ampere** is _____.

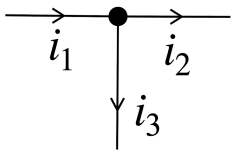


Fig. Q5.16.1

Q5.16.2 NAT 2016 1M

An air cored coil has a Q of 5 at a frequency of 100 kHz. The Q of the coil at 20 kHz (neglecting skin effect) will be _____.

Q5.16.3 NAT 2016 1M

In the circuit shown below $(v_1 + v_2) = [1\sin(2\pi 10000t) + 1\sin(2\pi 30000t)]V$. The RMS value of the current through the resistor R will be minimum if the value of the capacitor C in **microfarad** is _____.

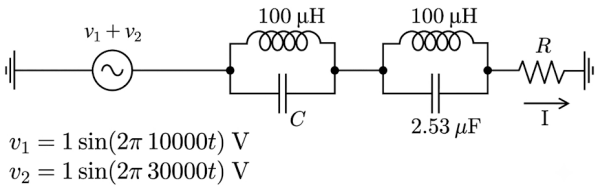


Fig. Q5.16.3

Q5.16.4 NAT 2016 2M

In the circuit shown below, $V_s = 101\angle 0\,V$, $R = 10\,\Omega$ and $\omega L = 100\,\Omega$.The current I_s is in phase with V_s. The magnitude of I_s in **milliampere** is _____.

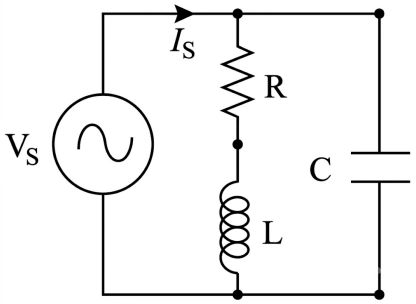


Fig. Q5.16.4

Q5.15.1 NAT 2015 2M

The circuit shown in the figure is in series resonance at frequency f_c Hz. The value of V_c is volts is _____ V.

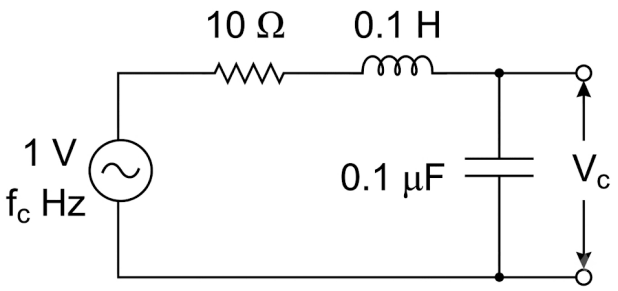


Fig. Q5.15.1

Q5.15.2 NAT 2015 2M

The output frequency of an LC tank oscillator employing a capacitive sensor acting as the capacitor of the tank is 100 kHz. If the sensor capacitance increases by 10 %, the output frequency is kilo-hertz becomes _____ kHz.

Q5.14.1 NAT 2014 1M

The average real power in watts delivered to a load impedance $Z_L = (4-j2)\,\Omega$ by an ideal current source $i(t) = 4\sin(\omega t + 20°)$ A is _____.

Q5.14.2 MCQ 2014 1M

Time domain expressions for the voltage $v_1(t)$ and $v_2(t)$ are given as $v_1(t) = V_m\sin(10t - 130°)$ and $v_2(t) = V_m\cos(10t + 10°)$.

Which one of the following statements is TRUE?

A. $v_1(t)$ leads $v_2(t)$ by 130°

B. $v_1(t)$ lags $v_2(t)$ by 130°

C. $v_1(t)$ lags $v_2(t)$ by $-130°$

D. $v_1(t)$ leadss $v_2(t)$ by $-130°$

Q5.14.3 NAT 2014 2M

A capacitor 'C' is to be connected across the terminals 'A' and 'B' as shown in the figure so that the power factor of the parallel combination becomes unity. The value of the capacitance required in μF is _____.

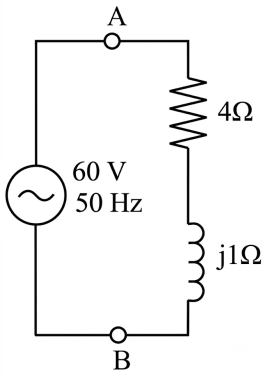


Fig. Q5.14.3

Q5.13.1 MCQ 2013 1M

A source $v_s(t) = V\cos 100\pi t$ has an internal impedance of $4 + j3\ \Omega$. If a purely resistive load connected to this source has to extract the maximum power out of the source, its value in Ω should be

A. 3 **B.** 4 **C.** 5 **D.** 7

Q5.13.2 MCQ 2013 2M

Two magnetically uncoupled inductive coils have Q factors q_1 and q_2 at the chosen operating frequency. Their respective resistances are R_1 and R_2. When connected in series, the effective Q factor of the series combination at the same operating frequency is

A. $q_1 + q_2$

B. $(1/q_1) + (1/q_2)$

C. $(q_1R_1 + q_2R_2)/(R_1 + R_2)$

D. $(q_1R_2 + q_2R_1)/(R_1 + R_2)$

Q5.13.3 MCQ 2013 2M

In the circuit shown below, if the source voltage $100\angle 53.13°$ V then the Thevenin's equivalent voltage in Volts as seen by the load resistance R_L is

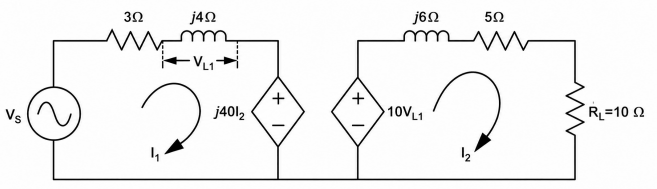


Fig. Q5.13.3

A. $100\angle 90°$ **B.** $800\angle 0°$
C. $800\angle 90°$ **D.** $100\angle 60°$

Q5.12.1 MCQ 2012 1M

The average power delivered to an impedance (4-j3) Ω by a current $5\cos(100\pi t + 100)$ A is

A. 44.2W **B.** 50W **C.** 62.5W **D.** 125W

Q5.12.2 MCQ 2012 1M

In the circuit shown below, the current through the inductor is

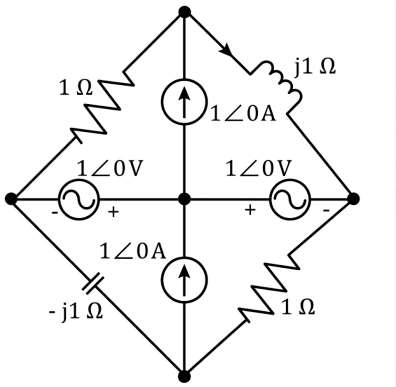


Fig. Q5.12.2

A. $\frac{2}{1+j}$A **B.** $\frac{-1}{1+j}$A

C. $\frac{1}{1+j}$A **D.** 0 A

Q5.12.3 MCQ 2012 2M

Assuming both the voltage sources are in phase, the value of R for which maximum power is transferred from circuit A to circuit B is

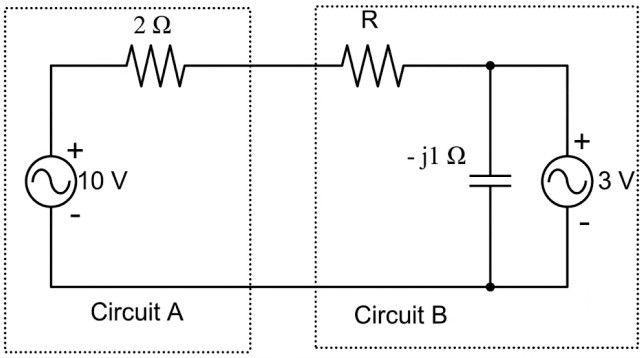


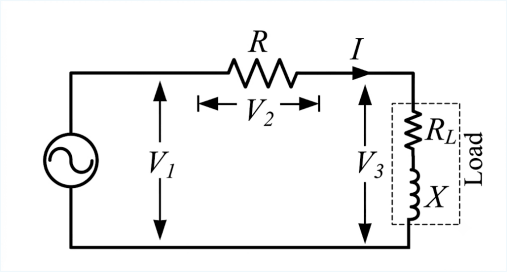
Fig. Q5.12.3

A. 0.8Ω **B.** 1.4Ω **C.** 2Ω **D.** 2.8Ω

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions 5.12.4 and 5.12.5:
In the circuit shown, the three voltmeter readings are $V_1 \cong 220V$, $V_2 \cong 122V$, $V_3 \cong 136V$.



Q5.12.4 MCQ 2012 2M

The power factor of the load is

A. 0.45 **B.** 0.50 **C.** 0.55 **D.** 0.60

Q5.12.5 MCQ 2012 2M

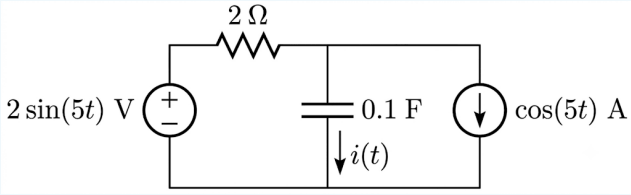
If $R_L \cong 5\Omega$, the approximate power consumption in the load is

A. 700W **B.** 750W **C.** 800W **D.** 850W

Common Data Questions

The next 2 questions share the data below.

Common Data for Questions 5.11.1 and 5.11.2: Consider the circuit shown below.



Q5.11.1 MCQ 2011 2M

The current i(t) through the capacitor is

A. sin (5t) A **B.** cos (5t) A

C. $\sin(5t - 45°)$ A **D.** 1 A

Q5.11.2 MCQ 2011 2M

The average total power delivered by the two sources is

A. 0 W **B.** 0.5 W **C.** 2 W **D.** 4 W

Common Data Questions

The next 2 questions share the data below.

Statement for Linked Answer Questions 5.10.1 an 5.10.2 : A coil having an inductance (L) of 10 mH and resistance R is connected in series with an ideal 100μF capacitor (C). When excited by a voltage source of value $10\sqrt{2}\cos(1000t)$ V, the series RLC circuit draws 20 W of power.

Q5.10.1 MCQ 2010 2M

The value of the coil resistance R is

A. 1Ω **B.** 2Ω **C.** 4Ω **D.** 5Ω

Q5.10.2 MCQ 2010 2M

The Q factor of the coil at an angular frequency of 1000 rad/s is

A. 1 **B.** 2 **C.** 4 **D.** 5

Q5.08.1 MCQ 2008 2M

In the circuit shown below the maximum power that can be transferred to the load Z_L is

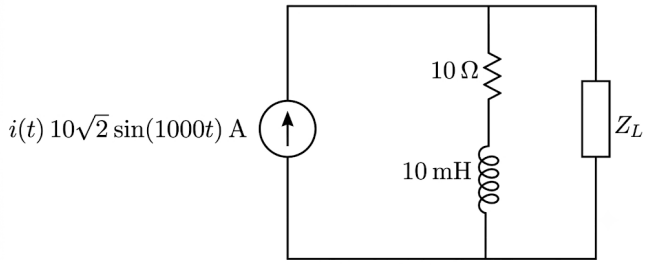


Fig. Q5.08.1

A. 250 W **B.** 500 W
C. 1000 W **D.** 2000 W

Q5.08.2 MCQ 2008 2M

In the circuit shown below the average power consumed by the 1 Ω resistor is

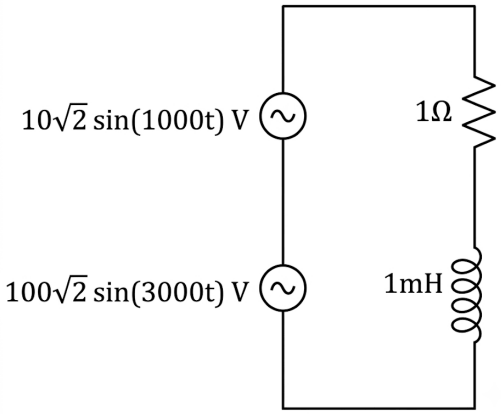


Fig. Q5.08.2

A. 50W
B. 1050W
C. 5000W
D. 10100W

Q5.08.3 MCQ 2008 2M

For the circuit shown below the voltage across the capacitor is

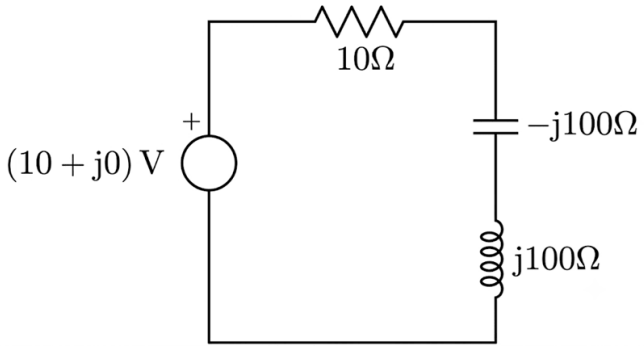


Fig. Q5.08.3

A. (10+j0)V
B. (100+j0)V
C. (0+j100)V
D. (0-j100)V

Q5.08.4 MCQ 2008 2M

For the circuit shown below the steady-state current I is

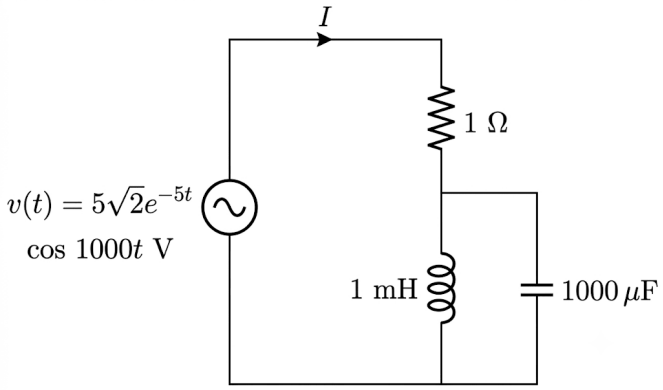


Fig. Q5.08.4

A. 0 A
B. $5\sqrt{2}\cos(1000t)$ A
C. $5\sqrt{2}\cos(1000t - \frac{\pi}{4})$ A
D. $5\sqrt{2}$ A

Q5.08.5 MCQ 2008 1M

If a current of $[-6\sqrt{2}\sin(100\pi t) + 6\sqrt{2}\cos(300\pi t + \frac{\pi}{4})$ A] is passed through a true RMS ammeter, the meter reading will be

A. $6\sqrt{2}A$ **B.** $\sqrt{126}A$ **C.** $12A$ **D.** $\sqrt{216}A$

Q5.07.1 MCQ 2007 2M

In the circuit shown in the figure, the input signal is $v_i(t) = 5 + 3\cos(\omega t)$. The steady state output is expressed at $v_o(t) = P + Q\cos(\omega t - \phi)$. If $\omega CR = 2$, the values of P and Q are

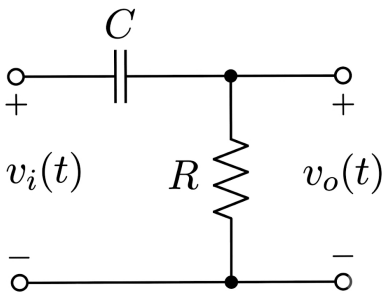


Fig. Q5.07.1

A. $P = 0$ and $Q = 6/\sqrt{5}$ **B.** $P = 0$ and $Q = 3/\sqrt{5}$
C. $P = 5$ and $Q = 6/\sqrt{5}$ **D.** $P = 5$ and $Q = 3$

Q5.07.2 MCQ 2007 2M

In the circuit shown in the following figure, the current through the 1 Ω resistor is

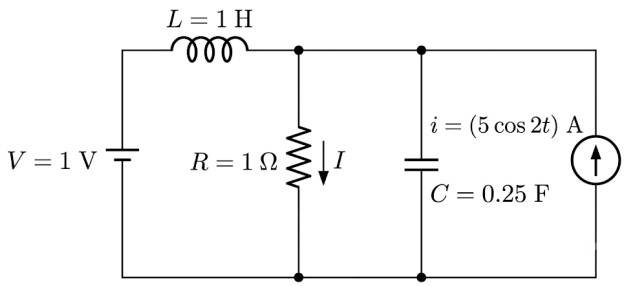


Fig. Q5.07.2

A. (1+5 cos 2t) A **B.** (5 + cos 2t) A
C. (1 - 5 cos 2t) A **D.** 6 A

Q5.06.1 MCQ 2006 1M

The root-mean-square value of a voltage waveform consisting of a superimposition of 2 V dc and a 4 V peak-to-peak square wave is

A. 2 V **B.** $\sqrt{6}$ V
C. $\sqrt{8}$ V **D.** $\sqrt{12}$ V

Q5.05.1 MCQ 2005 2M

Figure shows a circuit which has a coil of resistance R and inductance L. At resonance, the Q-factor of the coil is given by

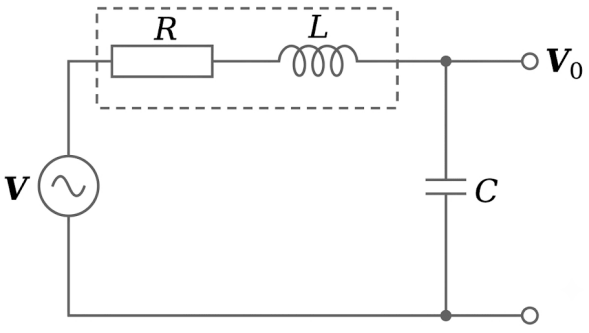


Fig. Q5.05.1

A. $\frac{V - V_0}{V}$ **B.** $\frac{V_0}{V}$
C. $\frac{V - V_0}{V_0}$ **D.** $\frac{V}{V_0}$

Q5.04.1 MCQ 2004 2M

An unknown impedance Z is shunted across an ac current source of value 2cos(10t) A. The current source develops a voltage of $6\sqrt{2}cos(10t + 45°)V$. The value of the impedance is

A. 3 Ω resistor in series with 0.3 H inductor

B. 3 Ω resistor in parallel with 0.3 H inductor

C. 3 Ω resistor in series with 0.3 F capacitor

D. 3 Ω resistor in parallel with 0.3 F capacitor

Q5.99.1 MCQ 1999 1M

For the waveform $V(t) = 2+\cos\left(\omega t + \frac{\pi}{6}\right)$ the ratio $V_{rms}/V_{average}$ is

A. $\frac{3}{2\sqrt{2}}$ **B.** $\sqrt{\frac{3}{2}}$
C. π **D.** $\frac{\pi}{2}$

Q5.98.1 MCQ 1998 2M

For the current in branch AB shown in figure, the voltage V_{in} is

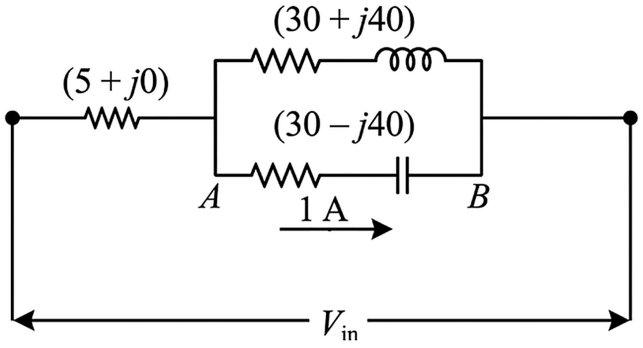


Fig. Q5.98.1

A. 55 V **B.** 110 V **C.** 56 V **D.** 90 V

Answer Key

Q5.26.1 D	Q5.26.2 25 to 25	Q5.25.1 A
Q5.25.2 590 to 610	Q5.24.1 200 to 200	Q5.24.2 27.0 to 30.0
Q5.23.1 1.40 to 1.50	Q5.23.2 99.0 to 101.0	Q5.22.1 6.22 to 6.42
Q5.22.2 0.8 to 0.8	Q5.21.1 39 to 40	Q5.21.2 30 to 30
Q5.20.1 5 to 5	Q5.20.2 1 to 1	Q5.20.3 1.7 to 1.8
Q5.20.4 115 to 116	Q5.19.1 B	Q5.19.2 20000 to 20000
Q5.18.1 A	Q5.18.2 1009 to 1011	Q5.18.3 66 to 68
Q5.17.1 50 to 50	Q5.17.2 1.75 to 1.96	Q5.17.3 6 to 7
Q5.16.1 5 to 5	Q5.16.2 1 to 1	Q5.16.3 0.280 to 0.283
Q5.16.4 99.5 to 100.5	Q5.15.1 98 to 102	Q5.15.2 95 to 96
Q5.14.1 32 to 32	Q5.14.2 A	Q5.14.3 186 to 188
Q5.13.1 C	Q5.13.2 C	Q5.13.3 C
Q5.12.1 B	Q5.12.2 C	Q5.12.3 A
Q5.12.4 A	Q5.12.5 B	Q5.11.1 A
Q5.11.2 A	Q5.10.1 D	Q5.10.2 B
Q5.08.1 B	Q5.08.2 B	Q5.08.3 D
Q5.08.4 A	Q5.08.5 A	Q5.07.1 A
Q5.07.2 A	Q5.06.1 C	Q5.05.1 B
Q5.04.1 A	Q5.99.1 A	Q5.98.1 C

UNIT

VI

Two Port Parameters

Topics

1. Z-Parameters
2. Y-Parameters, Reciprocity and Symmetry
3. H, G and ABCD Parameters
4. Series and Parallel Connection of Two Port Networks

PrepFusion

Unit Snapshot*

Stream: IN	**Questions:** 9	**MCQ:** 9	**MSQ:** 0	**NAT:** 0
1M: 3	**2M:** 6	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2017)–4M(2012)

**See preface for how Avg Weightage and Range are calculated.*

Q6.26.1 MCQ 2026 2M

A two-port network has z-parameters $z_{11} = z_{22} = 10\ \Omega$, and $z_{12} = z_{21} = 5\ \Omega$. The value of R_L such that maximum power is transferred to R_L is _____ Ω.

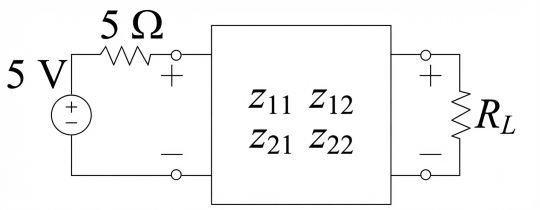


Fig. Q6.26.1

A. 5.67 **B.** 8.33 **C.** 10.33 **D.** 25.67

Q6.17.1 MCQ 2017 1M

The connection of two 2-port networks is shown in the figure. The $ABCD$ parameters of N_1 and N_2 networks are given as $\begin{bmatrix} A & B \\ C & D \end{bmatrix}_{N_1} = \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} A & B \\ C & D \end{bmatrix}_{N_2} = \begin{bmatrix} 1 & 0 \\ 0.2 & 1 \end{bmatrix}$

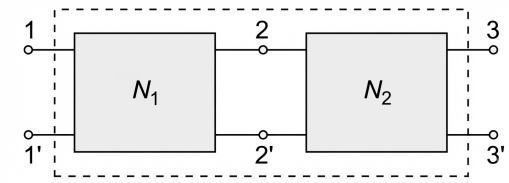


Fig. Q6.17.1

The $ABCD$ parameters of the combined 2-port network are

A. $\begin{bmatrix} 2 & 5 \\ 0.2 & 1 \end{bmatrix}$ **B.** $\begin{bmatrix} 1 & 2 \\ 0.5 & 1 \end{bmatrix}$

C. $\begin{bmatrix} 5 & 2 \\ 0.5 & 1 \end{bmatrix}$ **D.** $\begin{bmatrix} 1 & 2 \\ 0.5 & 5 \end{bmatrix}$

Q6.13.1 MCQ 2013 2M

Considering the transformer to be ideal, the transmission parameter 'A' of the 2-port network shown in the figure below is

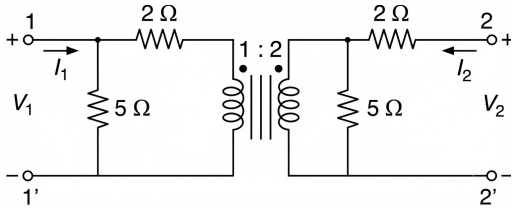


Fig. Q6.13.1

A. 1.3 **B.** 1.4 **C.** 0.5 **D.** 2.0

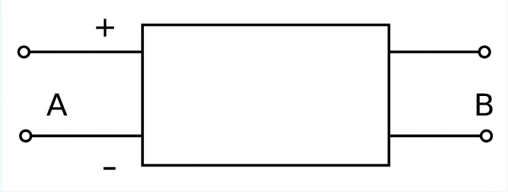
Common Data Questions

The next 2 questions share the data below.

Answer Questions Q6.12.1 and Q6.12.2 using the following data. With 10 V dc connected at port A in the linear nonreciprocal two-port network shown below, the following were observed:

i) 1 Ω connected at port B draws a current of 3 A

ii) 2.5 Ω connected at port B draws a current of 2 A



Q6.12.1 MCQ 2012 2M

With 10 V dc connected at port A, the current drawn by 7 Ω connected at port B is

A. 3/7A **B.** 5/7A **C.** 1A **D.** 9/7A

Q6.12.2 MCQ 2012 2M

For the same network, with 6 V dc connected at port A, 1 Ω connected at port B draws 7/3 A. If 8 V dc is connected to port A, the open circuit voltage at port B is

A. 6V **B.** 7V **C.** 8V **D.** 9V

Q6.08.1 MCQ 2008 2M

For the circuit shown below the input resistance $R_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0}$ is

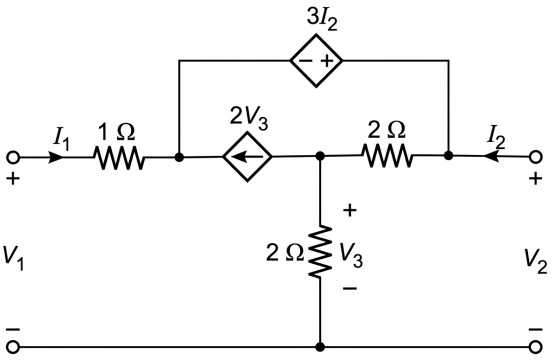


Fig. Q6.08.1

A. $-3\ \Omega$ **B.** $2\ \Omega$ **C.** $3\ \Omega$ **D.** $13\ \Omega$

Q6.07.1 MCQ 2007 1M

The DC voltage gain $\frac{V_o}{V_i}$ in the following circuit is given by

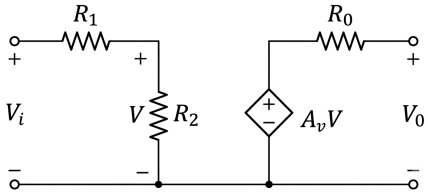


Fig. Q6.07.1

A. $A_v \dfrac{R_2}{R_1 + R_2}$ **B.** $A_v \dfrac{R_1}{R_1 + R_2}$

C. $A_v \dfrac{R_2}{R_1 + R_2} + R_0$ **D.** A_v

Q6.07.2 MCQ 2007 2M

Consider the coupled circuit shown below:

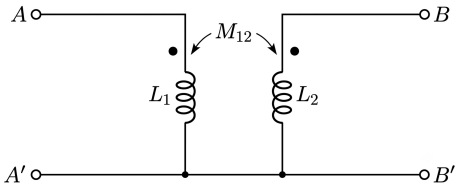


Fig. Q6.07.2

At angular frequency ω, this circuit can be represented by the equivalent T-network, shown below:

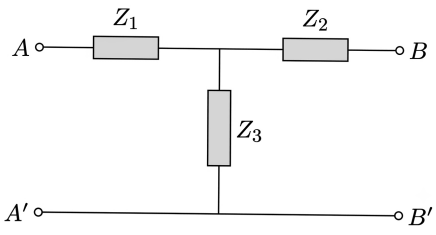


Fig. Q6.07.2

Indicate the correct set of expressions for the impedance of the T-network.

A. $Z_1 = j\omega(L_1 - M_{12}), Z_2 = j\omega(L_2 - M_{12}), Z_3 = j\omega M_{12}$

B. $Z_1 = j\omega(L_1 + M_{12}), Z_2 = j\omega(L_2 + M_{12}), Z_3 = j\omega M_{12}$

C. $Z_1 = j\omega L_1, Z_2 = j\omega L_2, Z_3 = -j\omega M_{12}$

D. $Z_1 = j\omega(L_1 - M_{12}), Z_2 = j\omega(L_2 - M_{12}), Z_3 = j\omega(L_1 + L_2 + M_{12})$

Q6.01.1 MCQ 2001 1M

Two identical 2-port networks with y-parameters $y_{11} = -y_{12} = -y_{21} = y_{22} = 1\text{S}$ are connected in cascade. The overall y-parameters will satisfy the condition

A. $y_{11} = 1\text{S}$ **B.** $y_{12} = -\frac{1}{2}\text{S}$

C. $y_{21} = -2\text{S}$ **D.** $y_{22} = 1\text{S}$

Answer Key

Q6.26.1 **B**	Q6.17.1 **A**	Q6.13.1 **A**
Q6.12.1 **C**	Q6.12.2 **C**	Q6.08.1 **D**
Q6.07.1 **A**	Q6.07.2 **A**	Q6.01.1 **B**

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